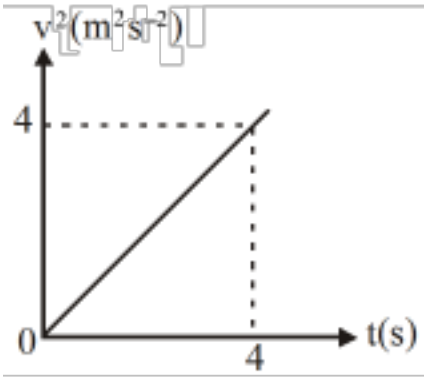
PHYSICS

1) A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at t = 4 s ?



(1) 4 m/s^2

(2) 1/4 m/s^2

(3) 1/2 m/s^2

(4) 0

2) Two bodies are thrown up at angles of 45° and 60° respectively with the horizontal. If both bodies attain same maximum height, then the ratio of speed with which these are thrown is :

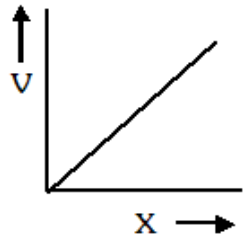
(1) $\sqrt{\frac{2}{3}}$

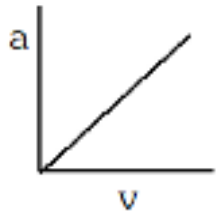
(2) $\frac{2}{\sqrt{3}}$

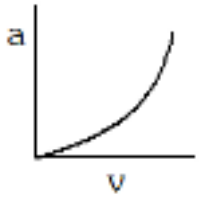
(3) $\sqrt{\frac{3}{2}}$

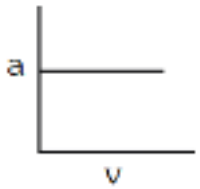
(4) $\frac{\sqrt{3}}{2}$

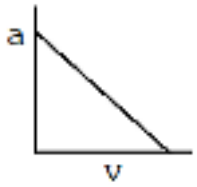
3) The velocity versus displacement graph of particle moving on straight line is given as below the corresponding acceleration versus velocity graph will be :



(1) 

(2) 

(3) 

(4) 

4) A ball is projected horizontally with velocity v_0 from the top of a tower. The horizontal distance on the ground is ηH, H is the height of tower. The speed of the ball when it reaches the ground is :-

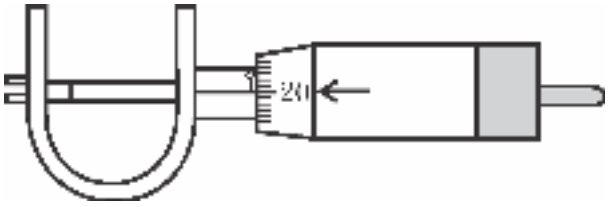
(1) $\eta\sqrt{\frac{gH}{2}}$

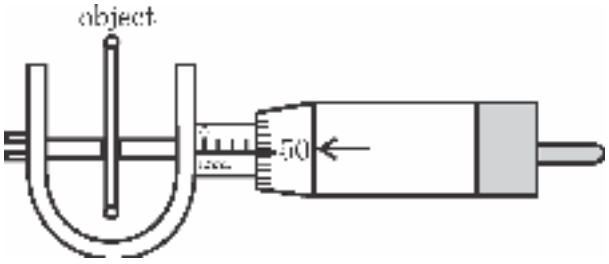
(2) $\sqrt{\frac{gH}{2\eta}}$

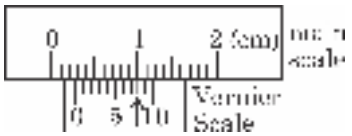
(3) $\sqrt{\left(\frac{\eta^2+4}{2}\right)gH}$

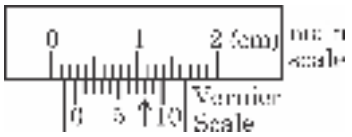
(4) $\sqrt{\left(\frac{2}{\eta^2+4}\right)gH}$

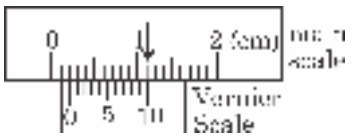
5) When a piece of wire is held diametrically in a screw gauge [pitch = 1 mm, number of division on circular scale = 100], the readings obtained are as shown. Now if we measure the same with help of vernier callipers [1 MSD = 1 mm, 10 divisions of vernier coinciding with 9 divisions of main scale] having a negative zero error of 0.5 mm, then find which of the following figures correctly represents the reading ?

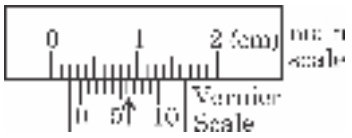




(1) 

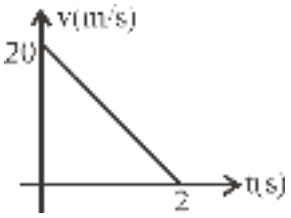
(2) 

(3) 

(4) 

6)

Velocity-time graph of a particle of mass 2 kg moving in a straight line is as shown in figure. Work done by all the forces on the particle is :-



(1) 400 J

(2) – 400 J

(3) – 200 J

(4) 200 J

7) A body is moved along a straight line by a machine delivering constant power. The distance moved by the body in time t is proportional to :

(1) $t^{1/2}$

(2) $t^{3/4}$

(3) $t^{3/2}$

(4) t^2

8) An annular ring with inner and outer radii R_1 and R_2 respectively is rolling without slipping with a uniform angular speed. The ratio of the forces experienced by the two particles situated on the inner

and outer parts of the ring, $\frac{F_1}{F_2}$ is

(1) 1

(2) $\frac{R_1}{R_2}$

(3) $\frac{R_2}{R_1}$

(4) $\left(\frac{R_1}{R_2}\right)^2$

9) The force required just to move a body up an inclined plane is double the force required just to prevent the body sliding down. If the coefficient of friction is 0.25, the angle of inclination of the plane is :-

(1) 37°

(2) 45°

(3) 30°

(4) 53°

10) A person is sitting in a travelling train and facing the engine. He tosses up a coin and the coin falls behind him. It can be concluded that the train is :

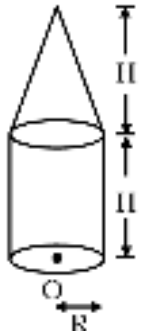
(1) Moving forward and gaining speed

(2) Moving forward and losing speed

(3) Moving forward with uniform speed

(4) Moving backward with uniform speed

11) If the density of material of cone and cylinder shown in figure is same, then which of the following is correct about the y-coordinate of COM (y_{COM}) w.r.t. point O.



(1) $y_{COM} > H$

(2) $y_{COM} = H$

(3) $y_{COM} < H$

(4) None of the above

12) Match the column:-

Column-I (Process of Collision)		Column-II (Coefficient of restitution)	
(a)	Ball dropped on a perfectly rigid floor, rebounds to same height	(1)	e = 0
(b)	Two clay balls collide and stick together	(2)	e = 1
(c)	A superelastic collision (fictitious, where KE increases)	(3)	0 < e < 1
(d)	A glass ball dropped on a steel plate, rebound to a lesser height	(4)	e > 1

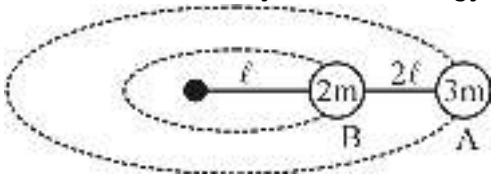
(1) a→2;b→1;c→4;d→3

(2) a→1;b→2;c→3;d→4

(3) a→2;b→1;c→3;d→4

(4) a→4;b→1;c→2;d→3

13) Two particle A and B are joined together as shown in the diagram and revolving on the horizontal circular path. Masses of A and B are 3m and 2m. P, V, E and F represent the linear momentum, velocity, kinetic energy and centripetal force respectively then match the column :



	Column-A		Column-B
(i)	$\frac{V_A}{V_B}$	(P)	$\frac{27}{2}$
(ii)	$\frac{E_A}{E_B}$	(Q)	$\frac{81}{4}$
(iii)	$\frac{F_A}{F_B}$	(R)	$\frac{9}{3}$
(iv)	$\left(\frac{P_A}{P_B}\right)^2$	(S)	$\frac{9}{2}$

(1) (i) → S, (ii) → Q, (iii) → P, (iv) → R

(2) (i) → Q, (ii) → P, (iii) → R, (iv) → S

(3) (i) → R, (ii) → P, (iii) → S, (iv) → Q

(4) (i) → P, (ii) → R, (iii) → Q, (iv) → S

14) Distance travelled by a particle moving in a circle of radius 8m is $x = 2t^2$, its acceleration after 2 sec will be in m/s^2 :-

(1) $\sqrt{24}$

(2) $4\sqrt{5}$

(3) 6

(4) $5\sqrt{4}$

15) One mole of Argon gas undergoes a process given by $PV^{3/2}$ = constant. If heat obtained by gas is Q and molar specific heat in the process in C, then which of the following is correct if temperature of gas changes by –26K? (R → gas constant)

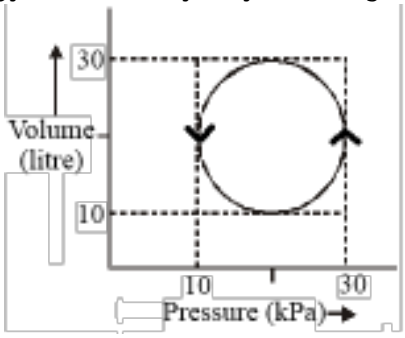
(1) C = 0.5 R, Q = 13R

(2) C = –0.5 R, Q = 15R

(3) C = –0.5 R, Q = 13R

(4) C = 0, Q = 13R

16) The heat energy absorbed by a system in going through a cyclic process shown in the adjacent figure is-



(1) 10^3 π J

(2) 10^2 π J

(3) 10^4 π J

(4) 10^7 π J

17) A water heater of power 2000 W is used to heat water. The specific heat capacity of water is 4200 J kg^{-1} K^{-1}. The efficiency of heater is 70%. Time required to heat 2 kg of water from 10°C to 60°C is :-

(1) 400 min

(2) 300 sec

(3) 200 sec

(4) 300 min

18) The ratio of the densities of the two bodies is 3 : 4 and the ratio of specific heats is 4 : 3 . Find the ratio of their thermal capacities for unit volume ?

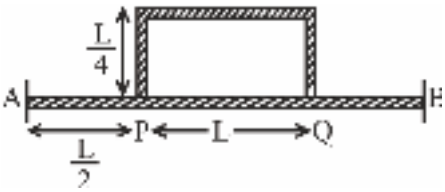
(1) 1 : 1

(2) 1 : 2

(3) 1 : 4

(4) 1 : 6

19) Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length 2L. Another bent rod PQ, of same cross-section and same material as AB and length $\frac{3L}{2}$, is connected across AB (See figure). In steady state, temperature difference between P and Q will be close to :



(1) 40°C

(2) 45°C

(3) 50°C

(4) 55°C

20) Two electric lamps A and B radiate the same power. Their filaments have the same dimensions, and have emissivities e_A and e_B. Their surface temperatures are T_A and T_B. The ratio $\frac{T_A}{T_B}$ will be equal to

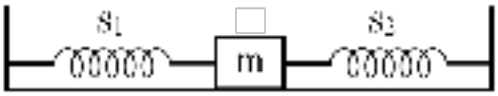
(1) $\left(\frac{e_B}{e_A}\right)^{1/4}$

(2) $\left(\frac{e_B}{e_A}\right)^{1/2}$

(3) $\left(\frac{e_A}{e_B}\right)^{1/2}$

(4) $\left(\frac{e_A}{e_B}\right)^{1/4}$

21) When a mass m is connected individually to two springs S_1 & S_2, the frequencies are f_1 & f_2. if both springs are attached as shown, the frequency of system will be –



(1) $f_1 + f_2$

(2) $\sqrt{f_1^2 + f_2^2}$

(3) $\left[\frac{1}{f_1} + \frac{1}{f_2}\right]^{-1}$

(4) $\sqrt{f_1^2 - f_1^2}$

22) The bob of a simple pendulum of length L is released at time t = 0 from a position of small angular displacement θ_0. Its linear displacement at time t is given by :

(1) $x = \theta_0 \sin 2\pi\sqrt{(L/g)}t$

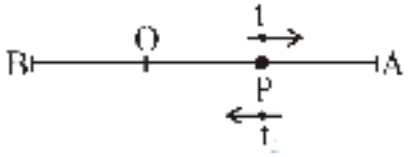
(2) $x = L\theta_0 \cos 2\pi\sqrt{(g/L)}t$

(3) $x = L\theta_0 \sin \sqrt{(g/L)}t$

(4) $x = L\theta_0 \cos \sqrt{(g/L)}t$

23) A particle executing SHM, starts from mean position. It passes through a certain point P at

time t_1 = 0.5 s and t_2 = 1.5 s with speed of 3 m/s. Find velocity at mean position :



(1) 3 m/s

(2) 5 m/s

(3) $3\sqrt{2}$ m/ s

(4) None

24) The equation of a stationary wave is Y = 10 sin $\frac{\pi x}{4}$ cos 20πt. The distance between two consecutive nodes in meters is -

(1) 4

(2) 2

(3) 5

(4) 8

25) A source of unknown frequency gives 4 beats/s, when sounded with a source of known frequency 500 Hz. The third harmonic of the source of unknown frequency gives 2 beats per second, when sounded with a source of frequency 1510 Hz. The unknown frequency is

(1) 510 Hz

(2) 504 Hz

(3) 496 Hz

(4) 490 Hz

26) Two waves having intensity I and 9I produce interference. If the resultant intensity at a point is 7I, what is the phase difference between the two waves :-

(1) 0°

(2) 60°

(3) 90°

(4) 120°

27) Match the column:

Column I		**Column II**	
A.	Sound waves	**P.**	Electromagnetic wave
B.	Waves in strings	**Q.**	Longitudinal wave
C.	Waves in a long spring	**R.**	Transverse wave
D.	Light waves	**S.**	Both transverse and longitudinal

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-R, C-S, D-P

(3) A-P, B-R, C-S, D-Q

(4) A-Q, B-P, C-R, D-S

28) Two liquids are at temperature 20°C and 40°C. When same mass of both of them is mixed then temperature of mixture is 32°C. What is ratio of their specific heat of liquids :-

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{1}{5}$

(4) $\frac{2}{5}$

29) If the speed of light (c), acceleration due to gravity (g) and pressure (p) are taken as the fundamental quantities, then the dimension of gravitational constant is :-

(1) $c^2g^0p^{-2}$

(2) $c^0g^2p^{-1}$

(3) cg^3p^{-2}

(4) $c^{-1}g^0p^{-1}$

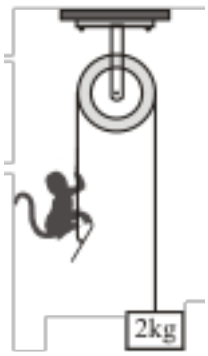
30) A new system of units is proposed in which unit of mass is α kg, unit of length is β m and unit of time is γ s. What will be value of 5J in this new system :

(1) $5\alpha\beta^2\gamma^{-2}$

(2) $5\alpha^{-1}\beta^{-2}\gamma^2$

(3) $5\alpha^{-2}\beta^{-1}\gamma^{-2}$

(4) $5\alpha^{-1}\beta^2\gamma^2$

31) 

A monkey of mass 1 kg is climbing on rope to just balance the weight of the block of mass 2kg. After some time monkey stop moving with respect to string. Find change in acceleration of monkey.

(1) g

(2) g/3

(3) 2g/3

(4) g/4

32) A body of mass 3kg is under a force which causes a displacement in it, given by $s = t^2/3$ (in m). Find the work done by the force in 2 second.

(1) 2 J

(2) 3.8 J

(3) 5.2 J

(4) 2.6 J

33) A rod of length L and mass M is bent to form a semicircular ring as shown in the adjoining



figure. The moment of inertia about XY is :-

(1) $\frac{ML^2}{2\pi^2}$

(2) $\frac{ML^2}{\pi^2}$

(3) $\frac{ML^2}{4\pi^2}$

(4) $\frac{2ML^2}{\pi^2}$

34) **Assertion (A) :** A couple does not exert a net force on an object even though it exerts a torque.
Reason (R) : Couple is a pair of two forces with equal magnitude but opposite directions acting simultaneously on a body in different lines of action.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

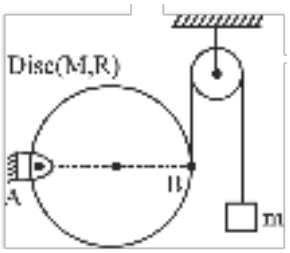
(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

35)

Find the value of mass m to keep the disc horizontal. (Given string is attached at point B on disc.)



(1) M

(2) $\frac{M}{2}$

(3) $\frac{M}{4}$

(4) $\frac{M}{3}$

36) The acceleration due to gravity g and mean density of earth ρ are related by which of the following relations ?
[G = gravitational constant and R = radius of earth] :

(1) $\rho = \frac{4\pi gR^2}{3G}$

(2) $\rho = \frac{4\pi gR^3}{3G}$

(3) $\rho = \frac{3g}{4\pi GR}$

(4) $\rho = \frac{3g}{4\pi GR^3}$

37) Potential energy of a 3kg body at the surface of a planet is –54J, then escape velocity will be:

(1) 18 m/s

(2) 162 m/s

(3) 36 m/s

(4) 6 m/s

38) Two solid spherical planets of equal radii R having masses 4M and 9M and their centres are separated by a distance 6R. A projectile of mass m is sent from the planet of mass 4 M towards the heavier planet. What is the distance r of the point from the lighter planet where the gravitational force on the projectile is zero ?

(1) 1.4 R

(2) 1.8 R

(3) 1.5 R

(4) 2.4 R

39) Young's modulus of steel is 2×10^{11} N/m^2. A steel wire has a length of 1 m and area of cross section 1 mm^2. The work required to increase its length by 1 mm is :-

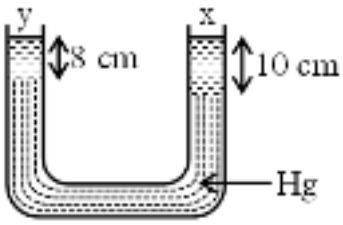
(1) 0.1 J

(2) 1 J

(3) 10 J

(4) 100 J

40) A liquid x of density 3.36 g/cm^3 is poured in a U tube which contains Hg. Another liquid y is poured in left arm with height 8 cm upper levels of x and y are same then density of y is:



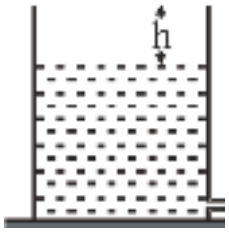
(1) 0.8 g/cm^3

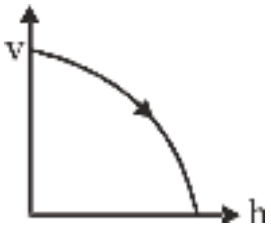
(2) 1.2 g/cm^3

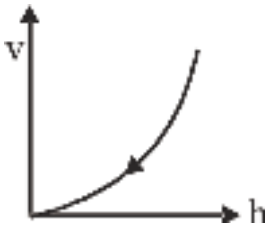
(3) 1.4 g/cm^3

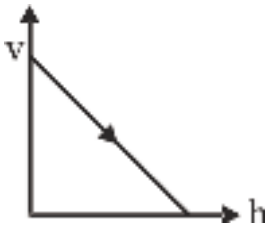
(4) 1.6 g/cm^3

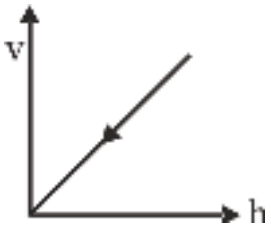
41) A rectangular tank is filled completely with water. A hole at its bottom is unplugged. The graph between the velocity of efflux (through a small hole) vs depth of water h from the top of tank.



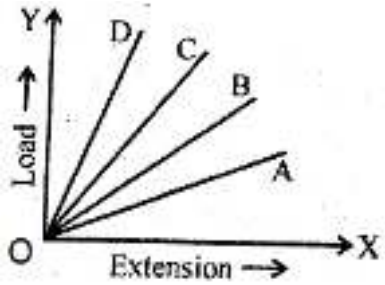
(1)


(2)


(3)


(4)


42) The load versus elongation graph for four wires of the same length and material is represented by the four lines OA, OB, OC and OD as shown in the figure. Which line represents the thinnest



wire?

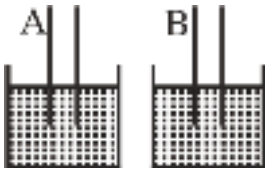
(1) line OB

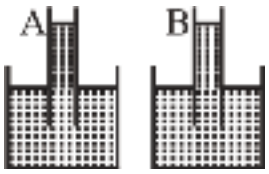
(2) line OA

(3) line OD

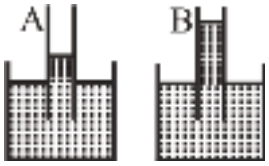
(4) line OC

43) A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

(1) 

(2) 

(3) 

(4) 

44) A vernier calliper has 10 division on the vernier scale which coincide with 9 division on the mainscale. The least count of the instrument is 0.1 mm. The value of one small division on main scale is :-

(1) 0.5 mm

(2) 2 mm

(3) 1 mm

(4) 0.1 mm

45) In a resonance tube, the first resonance with a tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is (in Hz) :-

(1) 500

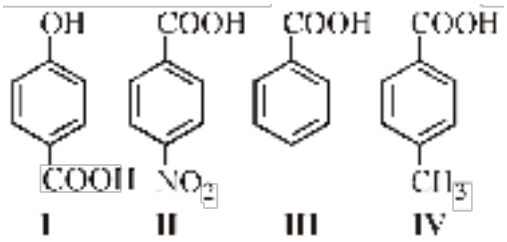
(2) 300

(3) 330

(4) 165

CHEMISTRY

1) The correct order of acid character of the following compounds is :



(1) III > II > I > IV

(2) IV > III > II > I

(3) I > II > III > IV

(4) II > III > IV > I

2) Which of the following compound is added to the sodium extract before addition of silver nitrate for testing of halogens ?

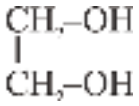
(1) Nitric acid

(2) Ammonia

(3) Hydrochloric acid

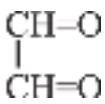
(4) Sodium hydroxide

3) $CH_3CHCl_2 \xrightarrow{2NaNH_2} X \xrightarrow[\Delta]{Fe} Y \xrightarrow[Zn/H_2O]{O_3} Z$ is :-

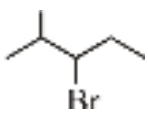
(1) 

(2) CH_3–CH_2–OH

(3) CH_3–COOH

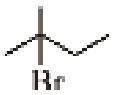
(4) 

4) Which alkyl bromide will produce only one alkene upon dehydrohalogenation :-

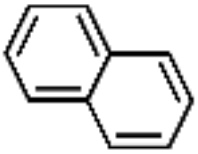
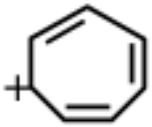
(1) 

(2) 

(3) 

(4) 

5) Match the Column I with Column II.

	Column-I		Column-II
(A)		(P)	Aromatic
(B)		(Q)	Non-aromatic
(C)		(R)	Anti-aromatic
(D)		(S)	Follows Huckel Rule

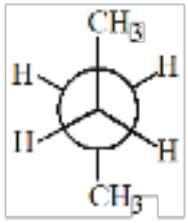
(1) (A)–Q ; (B)–P,S ; (C)–P,S ; (D)–R

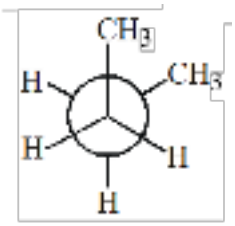
(2) (A)–P,S ; (B)–P,R ; (C)–P,S ; (D)–R,S

(3) (A)–Q,S ; (B)–R,S ; (C)–P,S ; (D)–Q,S

(4) (A)–P,S ; (B)–P,S ; (C)–R,S ; (D)–R

6) Which of the following best explains the reason for the relative stabilities of the conformers shown?

(I) 

(II) 

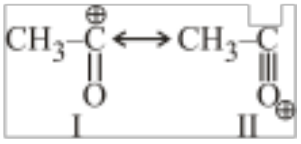
(1) I has more torsional strain

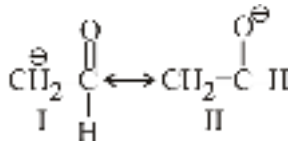
(2) I has more steric strain

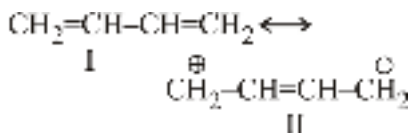
(3) II has more torsional strain

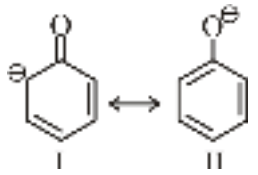
(4) II has more steric strain

7) In which pair's I more stable than II

(1) 

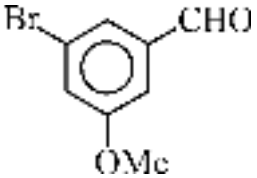
(2) 

(3) 

(4) 

8)

Choose the correct IUPAC name of compound :



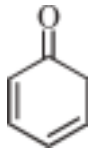
(1) 3-methyl-5-bromo benzene carbaldehyde

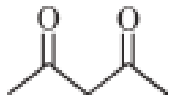
(2) 3-formyl-5-bromo phenyl methyl ether

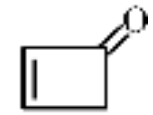
(3) 3-bromo-5-methoxy benzaldehyde

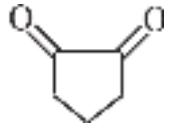
(4) 3-formyl-5-bromo-1-methoxy benzene

9) Which of following compound has keto content more than enol content :

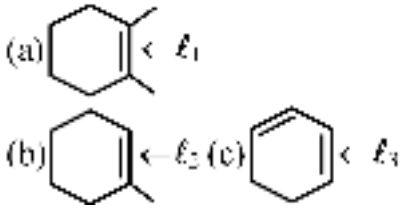
(1) 

(2) 

(3) 

(4) 

10) Correct order of indicated bond-length



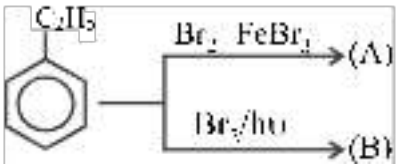
(1) $\ell_2 > \ell_3 > \ell_1$

(2) $\square_3 > \square_1 > \square_2$

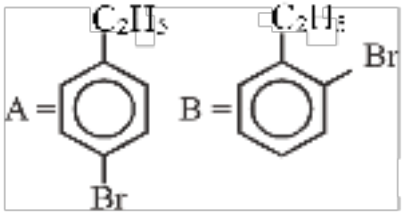
(3) $\square_3 > \square_2 > \square_1$

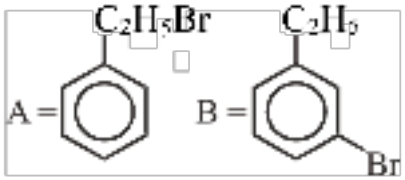
(4) $\square_1 > \square_3 > \square_2$

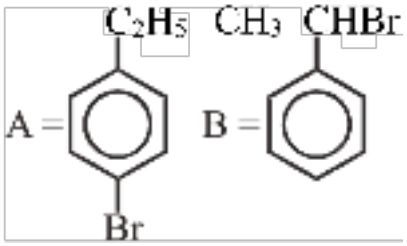
11)

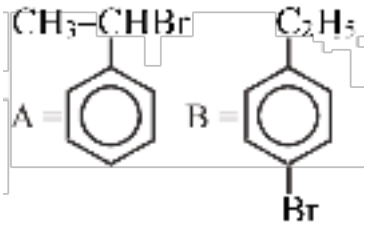


Identify product (A) & (B) :-

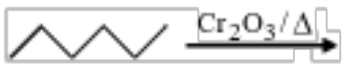
(1) 

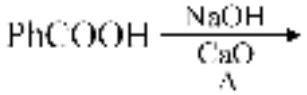
(2) 

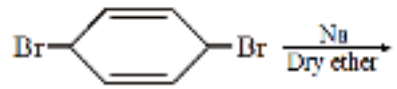
(3) 

(4) 

12) In which of the following reaction benzene is not formed as major product?

(1) 

(2) 

(3) 

(4) 

13) In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-

(1) 18.20

(2) 16.76

(3) 15.76

(4) 17.36

14) Match **List-I** with **List-II** :

	List-I		**List-II**
(a)	Separation of aniline-water mixture	(i)	Fractional distillation
(b)	Separation of aniline-chloroform mixture	(ii)	Distillation under reduced pressure
(c)	Separation of glycerol from spent-lye	(iii)	Distillation
(d)	Separation of different fractions of crude oil	(iv)	Steam distillation

Choose the **correct answer** from the options given below

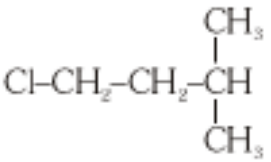
(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

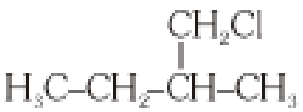
(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

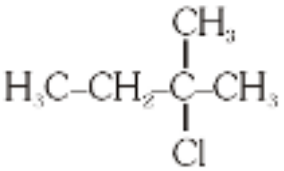
(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

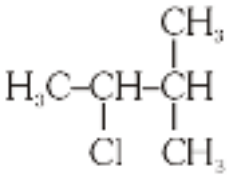
(4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

15) An alkene "A" on reaction with O_3 and Zn–H_2O gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is :-

(1) 

(2) 

(3) 

(4) 

16) Which of the following option is incorrectly matched according to given property.

(1) F < Cl < Br < I [Atomic radius]

(2) Cl > F > Br > I [Electron Gain Enthalpy]

(3) O > S > Se > Te > Po [Electron Gain Enthalpy]

(4) Li > Na > K > Rb > Cs [Ionisation Enthalpy]

17) Which of the following is the correct order of acidic, basic, amphoteric and neutral oxide respectively.

(1) As_2O_3, Na_2O, N_2O, Cl_2O_7

(2) Cl_2O_7, Na_2O, N_2O, As_2O_3

(3) N_2O, Cl_2O_7, As_2O_3, Na_2O

(4) Cl_2O_7, Na_2O, As_2O_3, N_2O

18) Which of the following is the incorrect statement regarding C_2 molecule by molecular orbital theory.

(1) It is diamagnetic

(2) Double bond in C_2 consists of both pi bonds

(3) Double bond in C_2 consists one sigma and one pi bond

(4) It has no unpaired electron

19) Boiling point of HF is greater than HCl due to presence of _____

(1) Intra molecular Hydrogen bonding

(2) Intermolecular Hydrogen bonding

(3) Vander Waal force of attraction

(4) Covalent bond

20) The shape of the following species are XeF_4, BrF_5, SF_4 respectively -

(1) Square planar, Square pyramidal, See-Saw

(2) Square pyramidal, See-Saw, Square planar

(3) Octahedral, Square planar, Trigonal bipyramidal

(4) Tetrahedral, Pyramidal, Trigonal bipyramidal

21) **Statement I :** A molecular orbital is polycentric
Statement II : The electron probability distribution around a group of nuceli is given by a molecular orbital

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

22) The correct order of increasing bond length of N = O, C = C, C = O, C = N is _____.

(1) C = O < C = C < C = N < N = O

(2) C = O < C = N < C = O < N = O

(3) C = O < N = O < C = C < C = N

(4) C = N < C = C < N = O < C = O

23) Which of the following statement is incorrect

(1) Noble gases have large positive electron gain enthalpy

(2) Electronegativity of any given element is not constant, it varies depending on the element to which it is bound.

(3) The process of adding electrons to the atom always endothermic

(4) The ionisation enthalpy is expressed in units of kJ/mole.

24) **Statement-I :** The hybridised orbitals are always equivalent in energy and shape.
Statement-II : The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

25) Which of the following represent incorrect match according to given property.

(1) $Na_2O < MgO < Al_2O_3$ [Lattice energy]

(2) AgF < AgCl < AgBr < AgI [Covalent character]

(3) $F_2 < Cl_2 < Br_2 < I_2$ [Boiling point]

(4) F–F > Cl–Cl > Br–Br [Bond dissociation energy]

26) Correct match the column-I with column-II and select the correct option given below.

	Column-I (Atomic number)		**Column-II (IUPAC official name)**
(A)	106	(p)	Nihonium
(B)	113	(q)	Tennessine
(C)	108	(r)	Seaborgium
(D)	117	(s)	Hassium

(1) (A - r) , (B - p), (C - q), (D - s)

(2) (A - r) , (B - p), (C - s), (D - q)

(3) (A - p) , (B - r), (C - s), (D - q)

(4) (A - s) , (B - p), (C - q), (D - r)

27) In H spectrum a certain electronic transition from an excited state to ground state in multistep produces 15 lines. Find number of lines in visible region.

(1) 5

(2) 4

(3) 3

(4) 2

28) **Assertion (A) :** At constant volume addition of inert gas has no effect on equilibrium.
Reason (R) : Addition of inert gas at constant volume does not change the partial pressure or the molar concentrations of the substances involved in the reaction.

(1) Both **(A)** & **(R)** are true and **(R)** is a correct explanation of the **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

29) Match Column-I with Column-II -

Column-I (Orbital)		**Column-II (Nodal properties)**	
(a)	5s	(P)	Number of radial node = 1
(b)	$3d_{yz}$	(Q)	Number of angular node = 0
(c)	$4d_{xy}$	(R)	Sum of radial and angular node = 3
(d)	$3p_x$	(S)	Number of radial node = 0

(1) (a) → (Q); (b) → (S); (c) → (P,R); (d) → (P)

(2) (a)→(Q,R); (b) → (Q,S); (c) → (S); (d) → (P)

(3) (a)→ (R,S); (b) → (P,Q); (c) → (R); (d) → (R)

(4) (a) → (P); (b) → (Q,R); (c) → (S); (d) → (R)

30) Which of the pair of two orbitals are located along the axis and not between the axis ?

(1) d_{xy}, d_{z^2}

(2) d_{xy}, p_z

(3) d_{yz}, p_x

(4) $p_z, d_{x^2-y^2}$

31) The value of stoichiometric coefficients (a, b, c, d and e) for the following reaction is :
$aPtCl_4 + bXeF_2 \rightarrow cPtF_6 + dClF + eXe$

(1) a = 2, b = 5, c = 2, d = 4, e = 5
(2) a = 1, b = 4, c = 1, d = 4, e = 4
(3) a = 1, b = 5, c = 1, d = 4, e = 5
(4) a = 1, b = 4, c = 1, d = 2, e = 4

32) The pair of compounds in which the metals are in their highest oxidation state is

(1) MnO_2, $FeCl_3$
(2) MnO_4^-, CrO_2Cl_2
(3) $MnCl_2$, $CrCl_3$
(4) $[NiCl_4]^{2-}$, Hg_2Cl_2

33) Which of the following is a redox reaction ?

(1) $NaCl + KNO_3 \rightarrow NaNO_3 + KCl$
(2) $CaCO_3 \rightarrow CaO + CO_2$
(3) $Zn + 2AgCN \rightarrow 2Ag + Zn(CN)_2$
(4) $Mg(OH)_2 + 2NH_4Cl \rightarrow MgCl_2 + 2NH_3 + 2H_2O$

34) For a reaction $2NO_2 + \frac{1}{2}O_2 \rightleftharpoons N_2O_5$, if equilibrium constant is k_p, then k_p for this reaction $2N_2O_5 \rightleftharpoons 4NO_2 + O_2$ will be:-

(1) k_p^2
(2) $\frac{2}{k_p}$
(3) $\frac{1}{(k_P)^2}$
(4) $\frac{1}{\sqrt{k_p}}$

35) $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$, the dissociation of 2 moles of CO_2 is carried out in container with volume 5 L. If at equilibrium 60% CO_2 is dissociated then total number of moles at equilibrium will be :-

(1) 2.3
(2) 2.6
(3) 3.0
(4) 3.2

36) If initially 1 mol each of A and B is taken in the reaction $A + B \rightleftharpoons C + D$, the degree of

dissociation would be :- [K is equilibrium constant]

(1) $\frac{\sqrt{K}}{\sqrt{K}+1}$

(2) $\sqrt{K}+1$

(3) $\frac{1-\sqrt{K}}{\sqrt{K}}$

(4) $\sqrt{K}-1$

37) 1 mole of each of $H_2(g)$ and $I_2(g)$ are heated in a closed vessel till equilibrium is obtained. If 80% of the hydrogen is converted to HI(g), the K_C at this temperature is :-

(1) 64

(2) 16

(3) 4

(4) 256

38)

For the reaction,
$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(\square)$,

ΔH = positive. At equilibrium which factor will not effect the concentration of NH_3 is.

(1) Change in pressure

(2) Change in volume

(3) Catalyst

(4) Change in temperature

39) 56 g of red hot iron is treated with 36 mL water. The weight of Fe_3O_4 produced would be (Atomic weight of Fe = 56)
$3Fe(s) + 4H_2O(\square) \rightarrow Fe_3O_4(s) + 4H_2\uparrow$

(1) 77.33 g

(2) 11.6 g

(3) 232 g

(4) 58 g

40) The mass percent of oxygen in NaOH is ?

(1) 40

(2) 16

(3) 8

(4) 1

41)

Match the following

	Reaction		Entropy change
(a)	A liquid vaporises	(i)	$\Delta S = 0$
(b)	Reaction is non spontaneous at all temperatures and ΔH is positive	(ii)	ΔS = Positive
(c)	Reversible adiabatic expansion of an ideal gas	(iii)	ΔS = negative

(1) (a) - ii, (b) - i, (c) - iii

(2) (a) - i, (b) - ii, (c) - iii

(3) (a) - iii, (b) - ii, (c) - i

(4) (a) - ii, (b) - iii, (c) - i

42) **Statement-I :** To some extent also determines the energy of the orbital in multi electron atom.
Statement-II : m_s refers to orientation of the spin of electron.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

43) **Statement (I) :** ΔG is known as free energy of reaction.
Statement (II) : ΔH_{sys} is the enthalpy change of reaction, $T\Delta S_{sys}$ is the energy which is not available to do useful work hence ΔG is net energy available to do useful work.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

44)

Which of the following reaction(s) represents enthalpy of atomization.
(a) $H_2(g) \rightarrow 2H(g)$
(b) $\frac{1}{3} NH_3(g) \rightarrow \frac{1}{3} N(g) + H(g)$
(c) $CH_4(g) \rightarrow C(g) + 4H(g)$
(d) $Na(s) \rightarrow Na(g)$
(e) $SO_2(g) \rightarrow S(s) + O_2(g)$

(1) a & c only

(2) a, c & d only

(3) a, b, c and d only

(4) a, b, c & e only

45) If $H^+ + OH^- \rightarrow H_2O + 13.7$ kCal, then heat of complete neutralisation of 1 gm molecules of H_2SO_4 with strong base in excess will be :

(1) –13.7 kCal

(2) – 27.4 kCal

(3) – 6.85 k Cal

(4) – 3.425 kCal

BIOLOGY

1) Which of the following statement is **correct** for transferases ?

(1) They catalyse oxidoreduction between two substrates.

(2) They are responsible for transfer of e^- and H^+.

(3) They catalyse a transfer of a group other than hydrogen between a pair of substrates.

(4) The example of these is aldolase.

2) _____ structure is the most stable structure of protein.

(1) Primary

(2) Secondary

(3) Tertiary

(4) Quaternary

3) **Statement (I)** : Alkaloid, flavonoids, rubber, essential oils etc are secondary metabolites.
Statement (II) : Nucleotides and amino acids are primary metabolites.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

4) Maltose has _____ glycosidic bond between α-D glucose & α-D glucose.

(1) 1' - 4''

(2) 1' - 2''

(3) 1' - 1''

(4) 4' - 2''

5) Identify incorrect statement from the options given below :

(1) Mesophyll cells of the leaves, have a large number of chloroplasts.

(2) Chloroplast align themselves along the walls of the mesophyll cells according to light intensity

(3) There is no division of labour within the chloroplast.

(4) Membrane system of chloroplast is responsible for trapping the light energy.

6) Match list-I with list-II

	List-I		List-II
A.	Synthesis of ATP and NADPH	i	Light reaction
B	Enzymatic reaction (synthesise sugar)	ii	In membrane system
C	Photochemical reactions	iii	In stroma
D	Dark reactions	iv	Not directly light driven

Choose correct answer from options given below :

(1) A - ii, B - iii, C - i, D - iv

(2) A - ii, B - iii, C - iv, D - i

(3) A - ii, B - iv, C - iii, D - i

(4) A - i, B - ii, C - iii, D - iv

7) Which of the following is/are green house crops ?

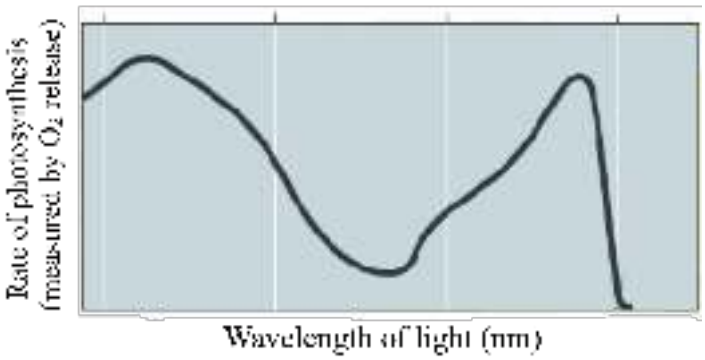
(1) Tomato

(2) Bell pepper

(3) Sugarcane

(4) Both (1) and (2)

8) Identify the below graph :



(1) Action spectrum of photosynthesis

(2) Absorption spectrum of chlorophyll a

(3) Absorption spectrum of chlorophyll b

(4) Action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a

9) Identify the correct statements :

A. Light reaction or photochemical phase includes light absorption, water splitting and oxygen release only.
B. Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase.
C. Cyclic photophosphorylation also occurs when light of wavelengths beyond 680 nm are available for excitation.
D. Peak of light absorption of the reaction center is same in both the photosystems.
Choose the **correct** answer from the options given below :-

(1) A and D only

(2) A, B and D only

(3) B and C only

(4) A, B and C only

10) In a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP. This reaction is related with :

(1) Glycolysis

(2) Tricarboxylic acid cycle

(3) Fermentation

(4) ETS

11) Identify the correct sequence of electron transport in ETS -

(1) Complex - I $\xrightarrow{e^-}$ complex -II

(2) Complex - II $\xrightarrow{e^-}$ complex - I

(3) Complex - II $\xrightarrow{e^-}$ complex - III

(4) Complex - III $\xrightarrow{e^-}$ complex - V

12) Identify the correct statement from following statements :-

(1) None of the intermediates in the respiratory pathway are utilised to synthesize any other compound.

(2) Neither photosynthesis nor respiration is an amphibolic pathway

(3) $\text{RQ} = \frac{\text{volume of } CO_2 \text{ evolved}}{\text{volume of } CO_2 \text{ consumed}}$

(4) Ultimately aerobic respiration, releases CO_2 & H_2O

13) Match list-I with list-II :

	List-I		**List-II**
A	Glucose → Glucose-6-Phosphate	I	Hexokinase
B	Fructose-6-Phosphate → Fructose-1,6-bisphosphate	II	Phosphofructokinase
C	Phosphoenol pyruvate → Pyruvate	III	Pyruvate kinase

D	Oxaloacetic acid + Acetyl coA $\xrightarrow{H_2O}$ Citric acid	IV	Citrate synthase

Choose the correct answer from the options given below :

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-II, B-III, C-I, D-IV

14) **Statement-I :-** During fermentation the reducing agent is NADH + H^+.
Statement-II :- NADH + H^+ is reoxidised to NAD^+ in both lactic acid and alcoholic fermentation.
Select the correct answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Only Statement I is correct

(3) Only Statement II is correct

(4) Both Statement I and Statement II are incorrect

15)

Match the following :

	Column-I		**Column-II**
A	GA_3	i	Stress Hormone
B	Kinetin	ii	Apical dominance
C	ABA	iii	Brewing industry
D	IBA	iv	Coconut milk

(1) A - i, B - ii, C - iii, D - iv

(2) A - ii, B - iii, C - iv, D - i

(3) A - iii, B - iv, C - i, D - ii

(4) A - iv, B - iii, C - ii, D - i

16) The volatile substance is released from ripened oranges that __A__ the ripening of stored un-ripened banana. It was identified as __B__.

(1) A-Reduced , B-Ascisic acid

(2) A-Hastened, B-Ethylene

(3) A-Stop, B-Auxin

(4) A-Hastened, B-Cytokinins

17) Ethylene is used to initiate flowering and for synchronising fruit set in :-

(1) Rice

(2) Potato

(3) Pineapples

(4) Cherry

18) How many of the following statement/s is/are correct?
(a) Cells differentiate and mature to perform specific functions is termed as differentiation.
(b) Interfascicular cambium and cork cambium formed from fully differentiated parenchyma cells.
(c) PGRs are small, simple molecules of diverse chemical composition.
(d) Only auxin breaks seed and bud dormancy, initiates germination in peanut seeds.

(1) Four

(2) Three

(3) Two

(4) One

19) Fill in the blank in given statement.
The cell of elongation zone has increased _A_ , cell enlargement and new _B_ deposition.
Choose correct answer from option given below.

(1) A. Vacuolation
B. Cell Wall

(2) A. Water
B. Cell Wall

(3) A. Vacuolation
B. Protoplasm

(4) A. Reserved food
B. Ergastic material

20) Kelps are related with :-

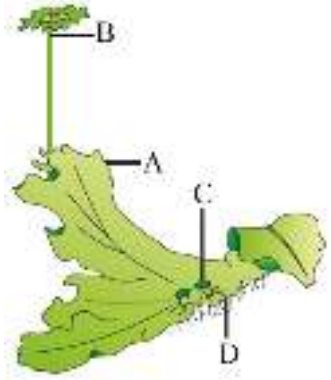
(1) Chlorophyceae

(2) Phaeophyceae

(3) Rhodophyceae

(4) Cyanophyceae

21) Examine the figure given below and choose the structure in which asexual buds are produced :-



(1) A

(2) B

(3) C

(4) D

22) Match the following genera with their respective divisions:-

(a)	*Ginkgo*	(p)	Angiospermae
(b)	*Sphagnum*	(q)	Gymnospermae
(c)	*Adiantum*	(r)	Bryophyta
(d)	*Eucalyptus*	(s)	Pteridophyta

	(a)	**(b)**	**(c)**	**(d)**
1	(p)	(q)	(r)	(s)
2	(s)	(p)	(q)	(r)
3	(q)	(s)	(r)	(p)
4	(q)	(r)	(s)	(p)

(1) 1

(2) 2

(3) 3

(4) 4

23) Which of the following is not a character related to red algae ?

(1) Sexual reproduction is oogamous.

(2) They occur in both well lighted areas and great depths of oceans.

(3) The stored food is cyanophycean starch, very similar to amylopectin and glycogen.

(4) They reproduce asexually by non motile spores.

24) Identify the set of correct statements.
A. The seed habit is considered an important step in evolution.
B. The prothallus of pteridophyta represents the gametophyte.
C. *Equisetum* is heterosporous pteridophyte.
D. The spread of living pteridophytes is limited and restricted to narrow geographical area.
E. The pteridophytes include horsetails and ferns.
Choose the correct answer from the options given below :

(1) C, D and E only

(2) A, B, C and D only

(3) B, C, D and E only

(4) A, B, D and E only

25) The development of pollen grain in gymnosperm take place with in __x__ retained on the parental __y__.

(1) x - Archegonia, y - gametophyte

(2) x - Antheridium, y - sporophyll

(3) x - Microsporangia, y - sporophyte

(4) x - Megasporangia, y - gametophyte

26) Which of the following group of organisms includes chemoautotrophs, according to five kingdom classification?

(1) Monera

(2) Fungi

(3) Plantae

(4) Animalia

27) Which one is an **incorrectly** matched pair ?

(1) Phycomycetes - *Mucor*, *Albugo*

(2) Ascomycetes - *Neurospora*, *Aspergillus*

(3) Basidiomycetes - *Puccinia*, *Agaricus*

(4) Deuteromycetes - *Ustilago*, *Trichoderma*

28) **Assertion :** A virus is a nucleoprotein and the genetic material is infectious.
Reason : No virus contains both RNA and DNA.

(1) **Assertion** and **Reason** both are correct but **Reason** is NOT correct explanation of **Assertion**.

(2) **Assertion** and **Reason** both are correct and **Reason** is correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

29) Identify the given figure and select the correct statements regarding it ?



(i) It is *Paramoecium*, having mixotrophic nutrition.
(ii) It is *Euglena* having one functional flagellum.
(iii) It is *Euglena* having pigments identical to higher plants.
(iv) It is *Paramoecium,* a ciliated protozoan.

(1) (i) & (iv)

(2) only (i)

(3) only (iii)

(4) (ii) & (iii)

30) Identify the mismatched one :-

(1)	Desmids	These are members of kingdom protista.
(2)	Slime moulds	Under suitable conditions these form plasmodium.
(3)	Ciliated protozoans	These possess two cillia-one large and another small.

(4)	Mycoplasma	These can survive without oxygen.

(1) 1

(2) 2

(3) 3

(4) 4

31) **Statement-I :** Sporozoans have an infectious spore like stage in their life cycle. The most notorious is *Plasmodium* which causes malaria, a disease which has a staggering effect on human population.
Statement-II : Slime moulds spores are dispersed by air currents.

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are incorrect.

(4) Both **Statement I** and **Statement II** are correct.

32) In which of the following organisms reproduction is not synonymous with growth ?

(1) Bacteria

(2) Unicellular algae

(3) *Amoeba*

(4) Housefly

33) Which of the following statement regarding nomenclature is correct?

(1) Generic name always begins with capital letter whereas specific epithet with small letter.

(2) Scientific name should be printed in italics.

(3) Scientific name when typed or handwritten should be underlined.

(4) All of the above

34) Arrange the following in ascending order of similar characteristics.
I. Family II. Genus
III. Class IV. Species

(1) Class < Family < Genus < Species

(2) Family < Class < Genus < Species

(3) Species < Order < Family < Class

(4) Class < Genus < Species < Family

35) Match the column-I with column-II and choose correct answer :

	Column-I		**Column-II**
(A)	Endoplasmic reticulum	(i)	Osmoregulation
(B)	Lysosome	(ii)	Digestion of protein
(C)	Golgi apparatus	(iii)	Steroidal hormone synthesis

(D)	Contractile vacuole	(iv)	Formation of glycolipid

(1) A-iii, B-ii, C-iv, D-i

(2) A-iv, B-iii, C-ii, D-i

(3) A-ii, B-iii, C-iv, D-i

(4) A-iii, B-iv, C-ii, D-i

36) Which of the following statement is incorrect regarding eukaryotic cell.

(1) The eukaryotes include all the protist, plant, animal and fungi.

(2) All eukaryotic cell are not identical.

(3) Eukaryotic cells possess an organised nucleus with a nuclear evelope.

(4) Ribosomes are non-membrane bound organelles found only in eukaryotic cells.

37) Plasmodesmata are :

(1) Cemented layers between the cells

(2) Locomotory structure

(3) Membrane connecting the nucleus with plasmalemna

(4) Connections between the cytoplasm of adjacent plant cells

38) Nucleoprotein fibers are called chromatin net. These are observed during :-

(1) Interphase nucleus

(2) Metaphase

(3) Anaphase

(4) Late prophase

39) The organelle considered as the site of aerobic respiration :-

(1) Produce cellular energy in the form of glucose

(2) Outer membrane forms number of infoldings known as cristae

(3) Divide by fission like division

(4) Matrix possesses single, circular and single stranded DNA molecule

40) Read the statements carefully :
(a) The ribosomes of a polysome translate the __A__ into proteins.
(b) ER divides the __B__ space into luminal (inside ER) and extra luminal (cytoplasm) compartments.
(c) Nucleolus is a site for active __C__ synthesis.
(d) Cilia and flagella are hair-like outgrowths of the __D__
Identify A,B,C and D respectively :

	(A)	**(B)**	**(C)**	**(D)**
(1)	tRNA	Intracellular	mRNA	cell wall
(2)	rRNA	Intracellular	tRNA	cell membrane
(3)	mRNA	Intracellular	rRNA	cell membrane

(4)	tRNA	Intercellular	mRNA	cell wall

(1) 1

(2) 2

(3) 3

(4) 4

41) If the amount of DNA in end of S-phase of onion root cell is 40 pg, what would be the amount of DNA in the cell at the time of entry in cell cycle at G_1 stage?

(1) 20 pg

(2) 40 pg

(3) 10 pg

(4) 5 pg

42) Identify the meiotic stage in which the sister chromatids separates :-

(1) Anaphase-I

(2) Anaphase-II

(3) Metaphase-I

(4) Metaphase-II

43) Read the following statements and find out the incorrect statement :-

(1) In an animal cell, cytokinesis is achieved by the appearance of a furrow in the plasma membrane.

(2) Furrow formation starts in the centre of cell and grows outward to meet the existing lateral walls.

(3) Cell plate represents the middle lamella between the walls of two adjacent plant cells.

(4) During cytokinesis, organelles like mitochondria and plastids get distributed between the two daughter cells.

44) Which one of the following cell division stage is correctly described with its event?

(1)	Anaphase	The bivalent chromosomes align on the equatorial plate.
(2)	Diakinesis	Beginning of dissolution of synaptonemal complex.
(3)	Metaphase	Spindle fibres attach to kinetochores of chromosomes.
(4)	Zygotene	Appearance of recombination nodule.

(1) 1

(2) 2

(3) 3

(4) 4

45) Which of the following option is not true about anaphase of mitosis and anaphase I meiosis ?

Anaphase of mitosis	**Anaphase of meiosis-I**

(1)	Centromere divides	Centromere does not divide
(2)	Each chromosome has single chromatid	Each chromosome has two chromatids
(3)	Total number of chromosome in a cell remains same	Total number of chromosome in a cell become double
(4)	Daughter chromosomes with single chromatids move towards the opposite poles	Chromosomes with two chromatids move towards the opposite poles

(1) 1

(2) 2

(3) 3

(4) 4

46) Which of the following sentence is incorrect regarding Hemichordata ?

(1) They have open type of circulatory system

(2) Respiration take place through Gills

(3) Fertilization is external in them

(4) Their excretory organ is Protonephridia

47) Identify the correct statements for bony fishes :-
(A) Body is not streamlined
(B) Skin is covered by placoid scales
(C) Air bladder and operculum present
(D) Mouth mostly terminal
(E) External fertilization

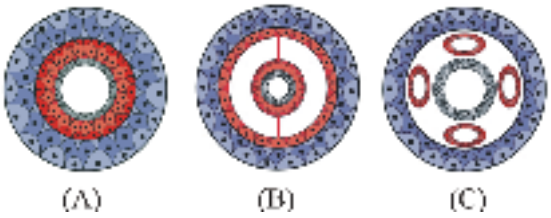
(1) A, B, D

(2) A, B, E

(3) C, D, E

(4) A, D, E

48) Which one of the following option is correct about the figures A, B and C given below?



(1) A-Acoelomate, B-Pseudocoelomate, C-Coelomate

(2) A-Coelomate, B-Acoelomate, C-Pseudocoelomate

(3) A-Pseudocoelomate, B-Coelomate, C-Acoelomate

(4) A-Acoelomate, B-Coelomate, C-Pseudocoelomate

49) Column A has some characters and B has their examples. Select the option, which correctly matches them :

Column-A		**Column-B**	
(P)	Canal system	(a)	*Asterias*
(Q)	Spiny body	(b)	*Obelia*
(R)	Stomochord	(c)	*Euspongia*
(S)	Metagenesis	(d)	*Columba*
(T)	Pneumatic bone	(e)	*Balanoglossus*

(1) P-c, Q-a, R-e, S-b, T-d

(2) P-a, Q-c, R-b, S-e, T-d

(3) P-e, Q-a, R-c, S-d, T-b

(4) P-b, Q-c, R-a, S-d, T-e

50) Identify the organism from the following hints :-
(A) Fresh water habitat with streamlined body
(B) Two chambered heart
(C) Air bladder present to regulate buoyancy
(D) Four pair of gills with operculum

(1) *Exocoetus*

(2) *Trygon*

(3) *Myxine*

(4) *Catla*

51) **Assertion :** Adult echinoderms are the radially symmetrical animals with true coelom.
Reason : Echinoderms have larval stage which is bilaterally symmetrical.

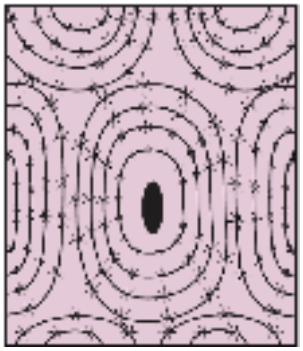
(1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion

(2) Both Assertion & Reason are True but the Reason is not the correct explanation of the Assertion

(3) Assertion is True statement but Reason is false

(4) Both Assertion and Reason are False statements

52) Select correct option with reference to ground substances of given diagram.



(1) Solid and pliable

(2) Semi-solid and non-pliable

(3) Abundant elastic fibres

(4) Rich in calcium salts and collagen fibres

53) How many statements are correct for simple cuboidal epithelium?
(A) The function is secretion and absorption.
(B) It is single layer of cube like cells.
(C) It is commonly present in duct of glands and tubular part of nephron.
(D) The nucleus is located at the base.

(1) 3

(2) 1

(3) 2

(4) 4

54) Which of following is not a part of male cockroach ?

(1) Titillator

(2) Phallomer

(3) Collateral glands

(4) Pseudopenis

55)

In cockroach, the sense organs are :-

(1) Anal style

(2) Spiracles

(3) Anal cerci

(4) Phallomeres

56) **Assertion :** In male frog, internal connection present between testes and kidney.
Reason : In frog, after spermatogenesis , sperms enter into kidney through vasa-efferentia.

(1) Assertion and reason both are true and the reason is correct explanation of assertion.

(2) Assertion and reason both are true but the reason is not correct explanation of assertion.

(3) Assertion is true but reason is wrong.

(4) Assertion and reason both are wrong.

57) Match the list I with list II given about structures of frog

List I		List II	
(i)	Copulatory pad	a	10–12
(ii)	Egg	b	10 pairs
(iii)	Vas efferentia	c	2500–3000
(iv)	Cranial Nerves	d	First digit

(1) i-b, ii-a, iii-c, iv-d

(2) i-d, ii-c, iii-a, iv-b

(3) i-d, ii-a, iii-c, iv-b

(4) i-a, ii-b, iii-c, iv-b

58) Persons with 'AB' blood group are called as "Universal recipient". This is due to

(1) Presence of anti-A and anti-B antibodies on RBCs

(2) Absence of anti-A and anti-B antibodies in plasma

(3) Absence of antigen A and B on the surface of RBCs

(4) Absence of antigen A and B in Plasma

59) Match the items given in Column-I with those in Column-II and select the correct option given below :

	Column-I		**Column-II**
(a)	Fibrinogen	(i)	Osmotic balance
(b)	Globulin	(ii)	Blood Clotting
(c)	Albumin	(iii)	Defence mechanism

(1) a – iii; b – ii; c – i

(2) a – i; b – ii; c – iii

(3) a – i; b – iii; c – ii

(4) a – ii; b – iii; c – i

60) Identify correct location of SAN (sino-atrial node) :-

(1) Right upper corner of left atrium.

(2) Right upper corner of right atrium.

(3) Left upper corner of left atrium.

(4) Left lower corner of right atrium.

61) **Assertion:** Angina is common among the middle age and elderly.
Reason: Acute chest pain appear when not enough oxygen in reaching the heart muscle.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

62) Which of the following statement is incorrect ?

(1) Due to atrial contraction flow of blood increases into vertricles by about 30 percent.

(2) The action potential is conducted to the atrial side by purkinje fibre.

(3) Ventricular systole causes the closure of tricuspid and bicuspid valve.

(4) When verticular pressure falls it causes the closure of semilunar valve.

63) Which one of the following is the correct statement for respiration in human beings ?

(1) Workers in grinding and stone-breaking industries may suffer from lung fibrosis

(2) About 90% of carbon dioxide (CO_2) is carried by heamoglobin as carbaminohaemoglobin

(3) Cigarette smoking does not cause any harm

(4) Neural signals from pneumotaxic centre in medulla region of the brain can increase the duration of inspiration

64) Choose the correct chemical equation related to respiration.

(1) $KHbO_2 \underset{RBC}{\rightleftharpoons} Hb + K H_2O$

(2) $Hb + O_2 \underset{\text{Dissociation in lungs}}{\overset{\text{Association in tissues}}{\rightleftharpoons}} HbO_2$

(3) $Na + HCO_3^- \underset{\text{Erythrocyte}}{\overset{\text{Plasma}}{\rightleftharpoons}} NaHCO_3$

(4) $HbO_2 \underset{\text{Association in lungs}}{\overset{\text{Dissociation in tissues}}{\rightleftharpoons}} Hb + O_2$

65) Which of the given option correctly represents total lung capacity (TLC) ?

(1) VC + RV

(2) IC + ERV

(3) ERV + RV

(4) IC + RV

66) **Statement-I :** Lungs are situated in the thoracic chamber which is anatomically an air tight chamber.
Statement-II : Any change is the volume of thoracic cavity is reflected in lungs cavity.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

67) **Statement - I** : Mechanism of breathing vary among different groups of animals depending mainly in their Habitats & Level of organisation.
Statement - II : Sponges respire through Gills.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I Incorrect But Statement II correct

68) Identify the correct option for the part of nephron which is lined by single layer of cube like cells with rounded nuclei in the centre along with minute finger like projections at free surface of the cells.

(1) Passive reabsorption of most of the glucose and amino acids

(2) Active reabsorption of nitrogenous wastes like ammonia

(3) Minimum reabsorption

(4) 70-80% of electrolytes and water are reabsorbed

69) The outline of principal events of urination in human are given below
(a) Stretch receptors on the wall of urinary bladder send signals to CNS
(b) The bladder fills with urine and become distended
(c) Micturition
(d) CNS passes motor messages to initiate the contraction of unstriped muscles of the bladder and relaxation of the urethral sphincter
Correct sequence of events is

(1) (a) → (b) → (c) → (d)

(2) (d) → (c) → (b) → (a)

(3) (b) → (a) → (d) → (c)

(4) (c) → (b) → (a) → (d)

70) Match the animals with their primary forms of nitrogenous waste excretion :

	Animals Group		**Excretory Waste**
(1)	Mammals	(a)	Ammonia
(2)	Birds	(b)	Urea
(3)	Bony fishes	(c)	Uric acid
(4)	Insects	(d)	Carbon dioxide

(1) 1-a, 2-b, 3-c, 4-d

(2) 1-b, 2-c, 3-a, 4-c

(3) 1-d, 2-c, 3-b, 4-a

(4) 1-c, 2-d, 3-b, 4-a

71) Give the below statement :-
Statement-I :- Some amount of urea may be retained in the kidney matrix of some animals to

maintain a desired osmolarity.
Statement-II :- Each kidney of an adult human measures 10-12 cm in width, 5-7 cm in length, 2-3 cm in thickness.
Option :-

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct & statement-II is incorrect.

(4) Statement-I is incorrect & statement-II is correct.

72) Which of the following sets of statements are correct ?
(A) JG cells release angiotensinogen in blood.
(B) ANF can cause vasodilation.
(C) Decrease in blood flow to the atria of the heart can cause the release of ANF.
(D) Aldosterone causes reabsorption of Na^+ and water from the distal parts of the tubule.
Choose the correct answer from the options given below :-

(1) A, B and C

(2) B and D

(3) B and C

(4) Only B

73)

The urine formed is a light yellow coloured and having pH ___A___. On an average, ___B___ of urea is excreted out per day.
Select the option which correctly represent A & B in the above mentioned statement.

(1) A-(6), B-(50-55 gm)

(2) A-(3), B-(20-25 gm)

(3) A-(6), B-(25-30 gm)

(4) A-(5), B-(15-20 gm)

74) The part of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc.

(1) Limbic system

(2) Brain stem

(3) Medulla

(4) Both (1) & (3)

75) Based on the number of axon and dendrites, the neurons are divided into three types
(i) Multipolar neuron found in __A__.
(ii) Bipolar neuron found in __B__.
(iii) Unipolar neuron found in __C__.

(1) A - Medulla oblongata ; B - Embryonic stage ;
C - Cranial nerve

(2) A - Cerebellum ; B - Embryonic stage ;
C - Dorsal root ganglion of spinal cord

(3) A - Cerebral cortex ; B - Retina ;
C - embryonic stage

(4) A - Pons ; B - Cerebral cortex ;
C - embryonic stage

76) Choose the correct statement from given below.

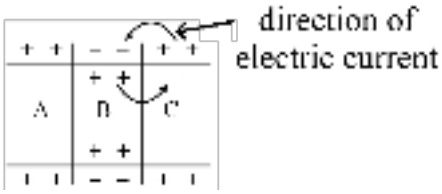
(1) Hypothalamus lies at the base of cerebellum

(2) Cerebral hemispheres are connected by a tract of nerve fibres called corpus Albicans.

(3) A large region in cerebral cortex that are neither clearly sensory nor motor called association area.

(4) Cerebrum wraps around pons.

77) In reference to given diagram, choose the correct set of option after reading following statements :-



a. The flow of impulse is from A to C
b. The flow of impulse is from C to A
c. A is in a state of polarisation
d. C is in a state of polarisation

(1) a, d

(2) a, c

(3) b, c

(4) b, d

78) The function of oestrogen hormone in human female is except :-

(1) Stimulation of growth and activities of female secondary sex organs.

(2) Development of growing ovarian follicles

(3) Appearance of female secondary sex characters

(4) Support Pregnancy

79) Match the hormone with its primary function :

Column-A		Column-B	
(1)	Estrogen	(A)	Stimulates ovarian follicle development
(2)	Progesterone	(B)	Supports pregnancy
(3)	PRL	(C)	Stimulates growth of female secondary sex organs
(4)	Follicle-stimulating hormone	(D)	Stimulates milk secretion

(1) (1)-(D), (2)-(C), (3)-(B), (4)-(A)

(2) (1)-(C), (2)-(B), (3)-(D), (4)-(A)

(3) (1)-(A), (2)-(B), (3)-(C), (4)-(D)

(4) (1)-(B), (2)-(A), (3)-(C), (4)-(D)

80) **Assertion :** Parathyroid hormone (PTH) increases blood calcium levels.
Reason : PTH acts on bones and stimulates bone resorption, releasing calcium into the bloodstream.

(1) Assertion is true but reason is wrong.

(2) Assertion and reason both are true but the reason is not correct explanation of assertion.

(3) Assertion and reason both are true and the reason is correct explanation of assertion.

(4) Assertion is and reason both are wrong.

81) Which of the following statements is correct ?

(1) Pars distalis produces six trophic hormones.

(2) Pineal gland secretes melanin, which plays a very Important role in the regulation of 24-hour (diurnal) rhythms of our body (e.g., rhythms of sleep and state of being awake, body temperature, etc.)

(3) The thyroid gland secrete thyrocalcitonin (TCT) which increases the blood Ca^{2+} levels and plays a major role in calcium homeostasis

(4) GIP stimulate gastric secretion and motility

82)

In humans ___A___ parathyroid glands are present on the ___B___ the thyroid gland-

(1) A - Two, B - Back

(2) A - Two, B - Front

(3) A - Four, B - Front

(4) A - Four, B - Back

83)

Which of the following is not a function of cortisol ?

(1) Hyperglycemia

(2) Stimulate inflammation

(3) Gluconeogenesis

(4) Stimulate RBC production

84) Find out correct statement regarding adrenal cortex hormone ?

(1) Underproduction of hormone alters carbohydrate metabolism causing acute weakness and fatigue.

(2) Adrenaline and nor-adrenaline are commonly called as catecholamines.

(3) Cortisol is involved in maintaining the hepatic portal circulation.

(4) Androgenic proteins play a role in the growth of axial hair, pubic hair and facial hair.

85) Identify A, B, C and D in the given statement.
Exopthalmic goitre is a form of __A__, characterised by enlargement of the __B__, protrusion of the eye ball, __C__ basal metabolic rate and weight __D__, also called Graves' disease.

(1) A-Hypothyroidism, B-thyroid gland, C-decreased, D-gain

(2) A-Hypothyroidism, B-Pituitary gland, C-increased, D-loss

(3) A-Hyperthyroidism, B-Pituitary gland, C-increased, D-gain

(4) A-Hyperthyroidism, B-Thyroid gland, C-increased, D-loss

86) Correctly match column-I with column-II.

	Column-I		**Column-II**
(A)	Amoeboid movement	(i)	Trachea
(B)	Ciliary movement	(ii)	Macrophage
(C)	Flagellar movement	(iii)	Limbs
(D)	Muscular movement	(iv)	Spermatozoa

(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

(2) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)

(3) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

87) How many of the statements given below are/is correct ?
(a) Muscles have special properties like excitability, contractility, extensibility and elasticity.
(b) Muscles are mesodermal in origin.
(c) 40-50% of the body weight of a human adult is contributed by blood

(1) One

(2) Two

(3) Three

(4) None is correct

88) Given below are two statements :
Statement-I : Activity of smooth muscles is not under voluntary control.
Statement-II : Smooth muscle assist in transportation of food through the digestive tract.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statemen-II is incorrect.

(3) Statement-I is incorrect but Statemen-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

89) Match the following columns.

	Column 1		**Column 2**
A	Sternum	I	Synovial fluid
B	Glenoid cavity	II	Vertebrae
C	Freely movable joint	III	Pectoral girdle
D	Cartilaginous joint	IV	Flat bones

Codes

	A	**B**	**C**	**D**
(1)	II	I	III	IV
(2)	IV	III	I	II
(3)	II	I	IV	III
(4)	IV	I	II	III

(1) 1

(2) 2

(3) 3

(4) 4

90) **Statement-I :** Our vertebral column formed by 26 serially arranged units called vertebra and is dorsally placed.
Statement-II : There are 10 pair of ribs. Each ribs is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	3	1	2	3	2	1	1	3	1	3	2	3	2	2	1	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	3	1	2	4	2	2	2	2	3	4	1	4	2	3	4	4	1	1
Q.	41	42	43	44	45															
A.	1	2	3	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	4	2	1	4	3	3	3	2	3	4	2	3	3	3	4	3	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	3	1	4	2	2	1	1	4	3	2	3	3	2	1	1	3	1	1
Q.	86	87	88	89	90															
A.	4	1	1	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	4	1	3	1	4	1	3	2	3	4	1	1	3	2	3	2	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	4	3	1	4	1	4	3	4	4	4	1	1	4	4	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	2	3	3	4	3	4	1	4	2	4	1	3	3	1	2	2	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	1	4	1	1	3	4	3	2	3	2	3	4	3	3	1	4	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	2	1	4	3	2	1	2	3										

SOLUTIONS

PHYSICS

1) $v^2 = t \Rightarrow v = t^{1/2} \Rightarrow a = \frac{dv}{dt} = \frac{1}{2}t^{-1/2}$

$\text{At } t = 4s,\ a = \frac{1}{2}(4)^{-1/2} = \frac{1}{4}m/s^2$

2) $H = \frac{u^2\sin^2\theta}{2g}$

Given $= \frac{u_1^2\sin^2 45^\circ}{2g} = \frac{u_2^2\sin^2 60^\circ}{2g}$

$\frac{u_1^2}{u_2^2} = \frac{\sin^2 60^\circ}{\sin^2 45^\circ}$

$\frac{u_1}{u_2} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}} = \sqrt{\frac{3}{2}}$

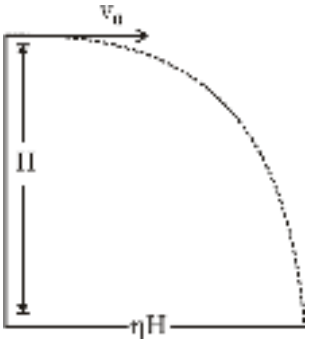
3) $v = kx$

$a = v\frac{dv}{dx} = (kx)(k)$

$a \propto x$

$a \propto \frac{v}{k}$

$a \propto v$

4) 

$x = \eta H$

$v_0 T = \eta H$

$v_0\sqrt{\frac{2H}{g}} = \eta H$

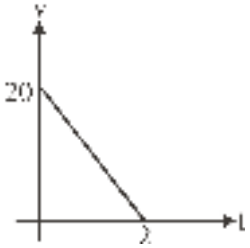
$v_0 = \eta\sqrt{\frac{gH}{2}}$

$v = \sqrt{v_0^2 + 2gH}$

$$= \sqrt{\eta^2 \frac{gH}{2} + 2gH}$$

$$v = \sqrt{\left(\frac{\eta^2 + 4}{2}\right) gH}$$

5) From screw guage
zero error = + 0.20 mm
Diameter of Chalk = 3.5 – 0.2
= 3.3 mm
for vernier calliper
zero error = – 0.05 cm
Thus it must show = 0.33 – 0.05
= 0.28 cm

6) 

Initial KE = $\frac{1}{2}mu^2$

$= \frac{1}{2} \times 2 \times (20)^2 = 400$

Final KE = 0 ⇒ Work done = – 400 J

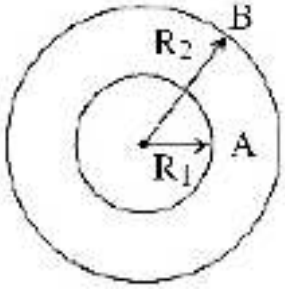
7) If power is constant then by using the relation P = F.v we can state that a.v = constant then using dv/dt = k/v we get $v = (2kt)^{1/2}$ then using $dx/dt = (2kt)^{1/2}$ on integrating we get displacement is directly proportional to $t^{3/2}$.

8) Since ω is constant, v would also be constant. So, no net force or torque is acting on ring. The force experienced by any particle is only along radial direction, or we can say the centripetal force.

The force experienced by inner part, $F_1 = m\omega^2 R_1$

The force experienced by outer part, $F_2 = m\omega^2 R_2$

$$\frac{F_1}{F_2} = \frac{R_1}{R_2}$$



9) Retardation in upward motion = g(sinθ + μcosθ)
 Force required just to move up F_{up} = mg (sinθ + μcosθ)
Similarly for down ward motion a = g(sinθ – μcosθ)
 Force required just to prevent the body sliding down F_{dn} = mg(sinθ – μcosθ)
According to problem $F_{up} = 2F_{dn}$
⇒ mg (sinθ + μcosθ) = 2mg (sinθ – μcosθ)
⇒ sinθ + μcosθ = 2 sinθ – 2μ cosθ
⇒ 3 μ cosθ = sinθ
⇒ tanθ = 3μ
⇒ $\theta = \tan^{-1}(3\mu) = \tan^{-1}(3 \times 0.25) = \tan^{-1}(0.75)$
$= 37°$

10) The coin falls behind him it means the velocity of train was increasing otherwise the coin fall directly into the hands of thrower.

11) Density of cone and cylinder is same
$M_{cylinder} = M_1 = \rho\pi R^2 H$

$$M_{cone} = M_2 = \frac{\rho\pi R^2 H}{3} = \frac{M_1}{3}$$

$$y_{COM} = \frac{M_1 y_1 + M_2 y_2}{M_1 + M_2}$$

$$= \frac{M_1 (H/2) + M_2 (H + H/4)}{M_1 + M_2}$$

$$y_{COM} = \frac{11H}{16} < H$$

12)

For perfect elastic e = 1
For perfect inelastic e = 0
For inelastic 0 < e < 1

13) Take same angular speed; linear speed v α r, Use $E = \frac{1}{2}mv^2$, $F = \frac{mv^2}{r}$, P = mv.
$\omega_A = \omega_B$

$$\frac{V_A}{V_B} = \frac{\omega_A r_A}{\omega_B r_B} = \frac{3\ell}{\ell} = 3 \rightarrow R$$

$$\frac{E_A}{E_B} = \frac{\frac{1}{2} \times 3mV_A^2}{\frac{1}{2} \times 2mV_B^2} = \frac{3}{2} \times (3)^2 = \frac{27}{2} \rightarrow P$$

$$\frac{F_A}{F_B} = \frac{3m\omega_A^2 r_A}{2m\omega_B^2 r_B} = \left(\frac{3}{2} \times \frac{1}{1} \times \frac{3}{1}\right) = \frac{9}{2} \rightarrow S$$

$$\left(\frac{P_A}{P_B}\right)^2 = \left(\frac{3m V_A}{2m V_B}\right)^2 = \left(\frac{3}{2} \times 3\right)^2 = \frac{81}{4} \rightarrow Q$$

14) Tangential acceleration from $v = \frac{dx}{dt}$ and centripetal acceleration $a_c = \frac{v^2}{r}$,

$x = 2t^2 \Rightarrow v = \frac{dx}{dt} = 4t$ At t = 2: v = 8m / s Tangential acceleration: $a_t = \frac{dv}{dt} = 4$ Centripetal acceleration: $a_c = \frac{v^2}{r} = \frac{64}{8} = 8$ Total acceleration:

$a = \sqrt{a_t^2 + a_c^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$

15) For polytropic process PV^m = constant

$C = C_v + \left(\frac{R}{1-m}\right)$ (Here $m = \frac{3}{2}$ and $C_v = \frac{3R}{2}$)

$\Rightarrow C = \frac{3R}{2} + \frac{R}{1-\frac{3}{2}} = \frac{3R}{2} - 2R = -0.5\,R$

Q = nCΔT = 1 (–0.5 R) (–26) = + 13R

16)

Q_{abs} = W = Area of curve

= π[a] [b]

= π[10 × 10^{+3}] × [10 × 10^{-3}]

= 100 π J

17) P × t × η = msΔT

2000 × t × 0.7 = 2 × 1000 × 4.2 × 50

t = 300 s

18) $\frac{\rho_1}{\rho_2} = \frac{3}{4}, \frac{s_1}{s_2} = \frac{4}{3}$

$\frac{H_1}{H_2} = \frac{\rho_1}{\rho_2} \times \frac{s_1}{s_2} = \frac{3}{4} \times \frac{4}{3} = 1$

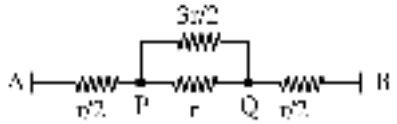
19)

$R = \frac{L}{KA} = r[R \propto L]$

$R_{AP} = \frac{r}{2}$; $R_{PQ\ (Straight)} = r$

$R_{PQ\ (Bend)} = \frac{3r}{2} = 1.5r$

$R_{QB} = \frac{r}{2}$



$R_{eq(PQ)} = \frac{R_1 R_2}{R_1 + R_2} = \frac{\left(\frac{3r}{2}\right)(r)}{\frac{3r}{2} + r} = \frac{3r}{5}$

$R_{eq(AB)} = \frac{r}{2} + \frac{3r}{5} + \frac{r}{2} = 1.6r$

$\because \Delta T \propto R \quad [V = IR]$

$\frac{\Delta T_{(PQ)}}{\Delta T_{(AB)}} = \frac{R_{PQ}}{R_{AB}} = \frac{(3r/5)}{(1.6r)} = \frac{3}{8}$

$\Delta T_{PQ} = \frac{3}{8}\Delta T_{AB} = \frac{3}{2} \times 120$

$\Delta T_{PQ} = 45°C$

20) The power radiated by a filament is $P = e\sigma AT^4$
where e = emissivity, σ = Stefan's constant,
T= surface temperature. or $e_A T_A^4 = e_B T_B^4$

$\Rightarrow \frac{T_A}{T_B} = \left(\frac{e_B}{e_A}\right)^{1/4}$

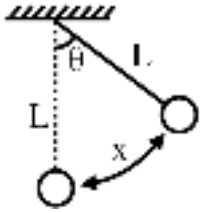
21) $f_1 = \frac{1}{2\pi}\sqrt{\frac{k_1}{m}}$ $f_2 = \frac{1}{2\pi}\sqrt{\frac{k_2}{m}}$

$f = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{m}}$

Solving we get $f = \sqrt{f_1^2 + f_2^2}$

22)

As pendulum starts from EP
So equation of SHM is



$\theta = \theta^0 \cos \omega t$ where $\omega = \frac{2\pi}{T}$

$\omega = \frac{2\pi}{2\pi\sqrt{\frac{L}{g}}} = \sqrt{\frac{g}{L}} \Rightarrow \theta = \theta_0 \cos\sqrt{\frac{g}{L}}t$

Linear displacement = x = Lθ

$x = L\theta^0 \cos\sqrt{\frac{g}{L}}t$

23)

T = 2(0.5 + 1.5)

T = 4 second, $\omega = \frac{2\pi}{4} = \frac{\pi}{2}$

Start M.P.

x = A sin ωt

$v = A\omega \cos \omega t$

$t = 0.5, v = 3$

$$3 = \frac{A\pi}{2} \cos\left(\frac{\pi}{2} \times 0.5\right)$$

$$A = \frac{6\sqrt{2}}{\pi}$$

M.P.

$V = A\omega$

$$V = \frac{6\sqrt{2}}{\pi} \times \frac{\pi}{2} = 3\sqrt{2} \text{ m/s}$$

24) $\frac{2\pi}{\lambda} = \frac{\pi}{4}$ $\lambda = 8$

N – N = $\frac{\lambda}{2} = \frac{8}{2}$ = 4m

25)

For f_1 = 504 Hz, third harmonic is 3 × 504 = 1512 Hz
For f_2 = 496Hz, third harmonic is 3 × 496 = 1488 Hz
Case-I : (1512 – 1510) = 2 b/s
Case-II : (1488 – 1510) = 22 b/s
So f_1 = 504 Hz

26) $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\phi$

Here, I_1=I, I_2=9I, and the resultant intensity I_R=7I.
Let's plug these values into the formula:

$$7I = I + 9I + 2\sqrt{I \cdot 9I} \cos\phi$$
$$7I = 10I + 2\sqrt{9I^2} \cos\phi$$
$$7I - 10I + 2(3I) \cos\phi$$
$$7I - 10I + 6I \cos\phi$$

Now, solve for cosϕ: −3I=6Icosϕ

$$\cos\phi = -\frac{3I}{6I} = -\frac{1}{2}$$

The angle ϕ for which cosϕ=−1/2 is **120°**.

27)

Sound wave → Longitudinal wave
Strings → Transverse wave
Long spring → Mixed
Light → EMW

28) Heat lost = Heat gain
ms_1 (32 – 20) = ms_2 (40 – 32)
$12s_1 = 8s_2$

$$\frac{s_1}{s_2} = \frac{2}{3}$$

29) Let $[G] \propto c^x g^y p^z$
by substituting the following dimensions
$[G] = [M^{-1}L^3T^{-2}]$, $[c] = [LT^{-1}]$, $[g] = [LT^{-2}]$
$[p] = [ML^{-1}T^{-2}]$
and by comparing the powers of both sides
we can get x = 0, y = 2, z = – 1
$\therefore [G] \propto c^0 g^2 p^{-1}$

30) $n_1u_1 = n_2u_2 \Rightarrow n_2 = n_1\frac{u_1}{u_2}$

$$n_2 = 5\left[\frac{5kg}{\alpha kg}\right]\left[\frac{m}{\beta m}\right]^2\left[\frac{sec}{\gamma\, sec}\right]^{-2}$$

$$J \rightarrow \frac{kg \times m^2}{sec^2}$$

31) to lift
T = 2g
$T - 1g = 1 \times a_1$
$\boxed{a_1 = g}$
After monkey stop
$$a_2 = \frac{(2-1)g}{2+1} = \frac{g}{3}$$
$$\Delta a = g - \frac{g}{3} = \frac{2g}{3}$$

32)

$$s = \frac{t^2}{3} \Rightarrow \frac{ds}{dt} = \frac{2t}{3}$$

i.e. $v = \frac{2t}{3}$
work done = ΔKE
$$WD = \frac{1}{2}m\left(v_f^2 - v_i^2\right)$$
at t = 0 $v_i = 0$
at t = 2 sec $v_f = \frac{4}{3}$ m/s
$$WD = \frac{1}{2} \times 3\left(\frac{16}{9} - 0\right)$$
$$= \frac{8}{3}J$$

33) MI of given ring = $\frac{MR^2}{2}$

Where $\pi R = L \Rightarrow R = \frac{L}{\pi}$

So, $I = \frac{ML^2}{2\pi^2}$

34)

Couple is a pair of two equal and opposite force having different lines of action.
So $F_{net} = 0$
But $\tau_{net} \neq 0$

35)

To keep the disc horizontal
Net torque = 0 (about point A)
Mg(R) – T(2R) = 0
MgR – mg (2R) = 0 (T = mg)

$$\boxed{m = \frac{M}{2}}$$

36) $g = \frac{GM}{R^2}$

$M = r \times \frac{4}{3}\pi R^3 \Rightarrow g = \frac{G \times 4\pi R^3 \times \rho}{3 \times R^2}$

$\Rightarrow \rho = \frac{3g}{4\pi GR}$

37)

$U = -\frac{GMm}{R}$

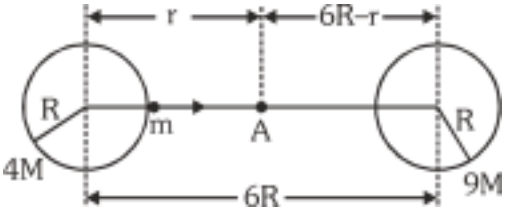
Escape velocity (v) = $\sqrt{\frac{2GM}{R}}$

$+54 = +\frac{GM}{R}(3)$

$\frac{GM}{R} = \frac{54}{3} = 18$

$v = \sqrt{2 \times 18} = 6$ m/s

38) The gravitational force on projectile of mass m at a distance r from a planet of mass 4M is zero



$$\frac{G(4M)m}{r^2} = \frac{G(9M)m}{(6R-r)^2}$$

$\Rightarrow 4(6R-r)^2 = 9r^2$

$\Rightarrow 2(6R - r) = 3r$

$\Rightarrow r = \frac{12}{5}R$

$\Rightarrow r = 2.4\ R.$

39) $W = U = \frac{1}{2}\frac{YA}{L}(\Delta L)^2$

$= \frac{1}{2} \times 2 \times 10^{11} \times 1 \times 10^{-6} \times \left(10^{-3}\right)^2$

$W = 0.1\ J$

40) For equilibrium

$8\ cm \times \rho_y \times g + 2\ cm \times \rho_{Hg} \times g = 10\ cm \times \rho_x \times g$

$8 \times \rho_y + 2 \times 13.6 = 10 \times 3.36$

$\rho_y = \frac{33.6 - 27.2}{8} = 0.8\ g/cm^3$

41) $v = \sqrt{2g(H-h)}$

42)

$F = \frac{YA}{\ell} \times \Delta\ell$

slope (m) = $\tan\theta = \frac{YA}{\ell}$

$\tan\theta \propto A$

43)

Surface tension of water is more than surface tension of soap solution.

44)

$0.1\ mm = \frac{10-9}{10}(MSD)$

$MSD = 1mm$

45) $\lambda = 2[0.49 - 0.16]$

$\lambda = 2[0.33]$

$n = \frac{v}{\lambda} = \frac{330}{2 \times 0.33} = 500\,Hz$

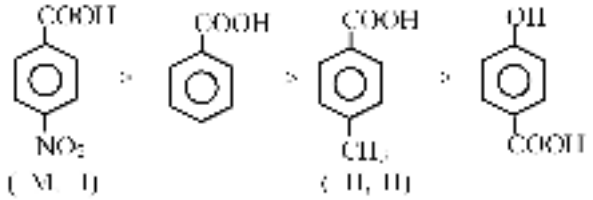
46) **Solution**:

Acidic strength $(K_a) \propto$ Stability of conjugate base.

$$\propto \frac{-M,\ -I}{+M,\ +I}$$

$\propto$ Delocalisation of charge in conjugate base.

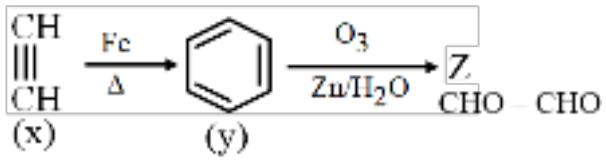
Overall acidic strength order:



Final Answer: (4)

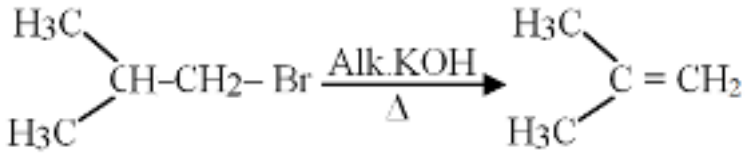
47) For testing of halogens, Nitric acid is added to the sodium extract because if CN^- or S^{2-} are present then they will be oxidised and removed before the test of halides.

48)



49)

Solution:



Only 1 alkene

Final Answer: Option (2)

50)

Cyclic, planar, completely
conjugated system follow Huckel rule
(4n + 2) π e^- rule) are aromatic

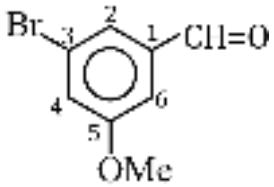
51) Gauche form is less stable than antiform because of more steric strain in gauche.

52)

neutral R.S. is more stable.

53)

Solution:

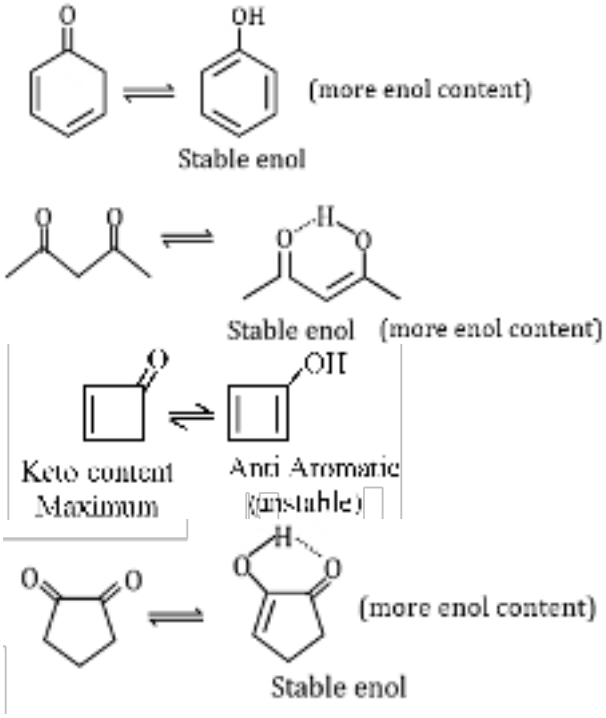


3-Bromo-5-methoxybenzaldehyde

Final Answer: (3)

54)

Solution:



Final Answer

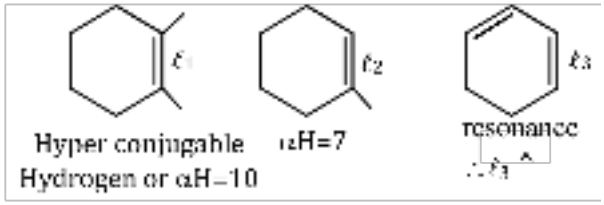
The correct option is (3)

55)

Solution:

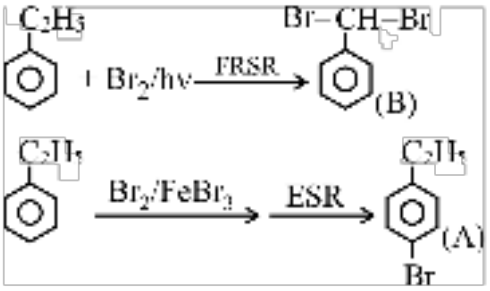
In alkenes C=C bond length increases due to hyperconjugation effect and resonance.

Bond length ∝ reso ∝ H-effect

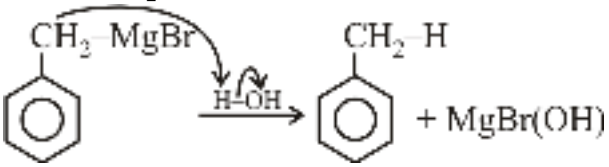


so bond length order is : $\ell_3 > \ell_1 > \ell_2$.

Final Answer:
The correct option is (2)

56) 

57) **Concept:**



Ans.(4)

58)

ω = 0.25 gm
V_1 = 40 mL
T_1 = 300 K
P_1 = (725 – 25) = 700 mm
N.T.P.
T_1 = 273 K
P_2 = 760 mm
V_2 = ?

$$V_2 = \frac{P_1V_1}{T_1} \times \frac{T_2}{P_2}$$

$$= \frac{700 \times 40}{300} \times \frac{273}{760}$$

= 33.52 ml

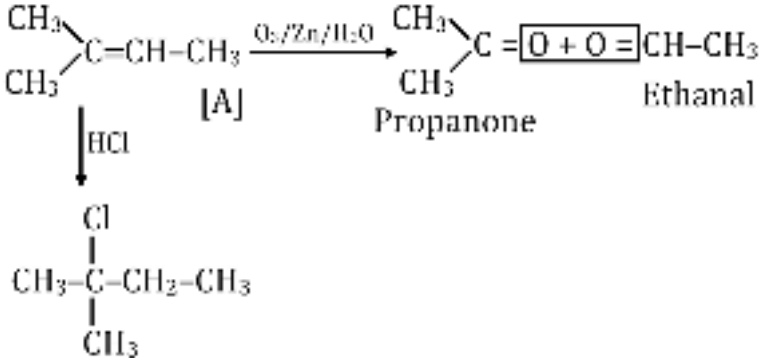
$$\% N = \frac{\text{Vol. at N.T.P.}}{8 \times \omega}$$

$$= \frac{33.52}{8 \times 0.25} = 16.76\%$$

59)

NCERT Fact

60)



61) NCERT-XI, Pg. # 90
$\Delta He_g \Rightarrow [S > Se > Te > Po > O]$

62) NCERT-XI, Pg. # 94
Cl_2O_7 Acidic
Na_2O Basic
As_2O_3 Amphoteric
N_2O Neutral

63) NCERT-11, Pg. # 130
$C_2 (12e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, [\pi(2p_x^2) = \pi(2P_y)^2]$
BO = 2[Both are pure π bond]
Because of the presence of four electrons in two pi molecular orbital

64) Due to presence of Inter molecular H-bond.

65) NCERT-XI, Pg. # 116
$XeF_4 \Rightarrow sp^3d^2$ [bp = 4, □p = 2]
$BrF_5 \Rightarrow sp^3d^2$ [bp = 5, □p = 1]
$SF_4 \Rightarrow sp^3d$ [bp = 4, □p = 1]

66)

NCERT-11, Chemical bonding M.O.T Postulates

67) NCERT-XI, Pg. # 108
C = O < N = O < C = C < C = N
121 122 133 138 pm

68) NCERT-XI, Pg. # 91
Addition of e^- in an atom may be endothermic or exothermic.

69) NCERT-XI, Pg. # 120

70) $\boxed{LE \propto |q_1 q_2|}$
[Size of Anion ↑ ⇒ Covalent character ↑]
[BP ∝ molecular weight ∝ Vander waal force of Attraction]
Bond strength ⇒ F–F < Cl–Cl > Br–Br

71) NCERT-XI, Pg. # 80
Z = 106 (Sg)
Z = 113 (Nh)
Z = 108 (Hs)
Z = 117 (Ts)

72) Total spectral line = $\frac{n(n-1)}{2} = 15$
⇒ $n^2 - n = 30$
or n = 6
⇒ lines in visible region = n – 2 = 4

73)

NCERT, Pg. # 211

74) This question is based on Angular nodes (ℓ), Radial nodes (n - ℓ - 1) Total nodes (n - 1)
(a) 5s (n = 5, ℓ = 0) has 0 angular nodes (ℓ = 0) This matches (Q).
(b) 3dyz (n = 3, ℓ = 2) has O radial nodes (n - ℓ - 1 = 3 - 2 - 1 = 0) This matches (S).
(c) 4dxy (n = 4, ℓ = 2) has 1 radial node (n - ℓ - 1 = 4 - 2 - 1 = 1)
and a total of 3 nodes (n - 1 = 4 - 1 = 3) . This matches (P) and (R).
(d) 3p (n = 3, ℓ = 1) has 1 radial node (n - ℓ - 1 = 3 - 1 - 1 = 1) This matches (P).
Therefore, the correct pairings are (a) → (Q), (b)→(S), (c) → (P,R), and (d) → (P).
Final Answer: (1)

75) NCERT, Pg.# 56

76) $PtCl_4 + 5XeF_2 \rightarrow PtF_6 + 4ClF + 5Xe$

77)

MnO_4^- , CrO_2Cl_2 → +7, +6
MnO_2, $FeCl_3$ → +4, +3
$MnCl_2$, $CrCl_3$ → +2, +3

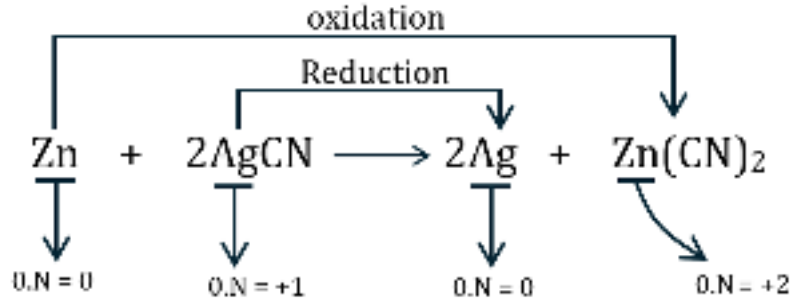
$[NiCl_4]^{2-}$, Hg_2Cl_2 → +2, +1

78)

Question Explanation : The question asks which reaction is redox reaction.

Concept : This question is based on Redox Reaction.

Solution :



This reaction is Redox Reaction
– Other reaction involve double displacement, decomposition or acid-base reactions-without oxidation state changes.

Final Answer: option (3)

79)

$2NO_2 + \frac{1}{2} O_2 \rightleftharpoons N_2O_5$; k_p

So $4NO_2 + O_2 \rightleftharpoons 2N_2O_5$; k_p^2

So $2N_2O_5 \rightleftharpoons 4NO_2 + O_2$; $\frac{1}{k_p^2}$

80) $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$ V = 5L

Initially	2	0	0
at Eq	2 – 1.2 = 0.8	1.2	0.6

$n_{eq} = 0.8 + 1.2 + 0.6$
$= 2.6$

81)

	A	+	B	⇌	C	+	D
initial	1		1		0		0
At eq^m	1 – x		1 – x		x		x

$k_{eq} = \frac{x^2}{(1-x)^2} \Rightarrow \frac{x}{1-x} = \sqrt{k}$

$\Rightarrow x = \frac{\sqrt{k}}{1+\sqrt{k}}$

Degree of dissociation = $\frac{x}{1} = x = \frac{\sqrt{k}}{1+\sqrt{k}}$

82) H_2 + I_2 ⇌ 2HI

1	1	0

1–α 1–α 2α
= 1–0.8 1–0.8 2 × 0.8
= 0.2 = 0.2 = 1.6
 $K_C = \frac{1.6 \times 1.6}{0.2 \times 0.2} = 64$

83)

Catalyst do not change position of equilibrium.

84) $3Fe_{(s)} + 4H_2O(\square) \rightarrow Fe_3O_4(s) + 4H_2 \uparrow$
3×56g 4 ×18 g 232 gm
assume 1mL water = 1g
L.R is Fe
 56 g Fe gives = $\frac{232}{3}$ g Fe_3O_4 = 77.33 g

85) % oxygen = $\frac{16}{40} \times 100$ = 40%

86)

(a) Liquid to gas → Entropy increases
(b) $\underbrace{\Delta G}_{\oplus} = \underbrace{\Delta H}_{\oplus} - \underbrace{T\Delta G}$ For ΔG to be +ve at all T, ΔS must be –ve
(c) Reversible process, ΔS = 0,

87)

NCERT, Pg. # 55

88)

NCERT, Pg. # 185

89)

Reaction (b) doesn't has one mole reactant in reaction (e) gaseous atoms are not formed.

90) Moles of H_2SO_4 = 1 mol
g eq. of H_2SO_4 = moles × V.F. = 1 × 2 = 2 g eq.
ΔH due to 2 g eq. = –13.7 × 2 = –27.4 kCal

BIOLOGY

91)

NCERT-XI, Pg. # 158

92) NCERT XI Pg : 111-112

93) NCERT-XI Pg. # 146

94)

Maltose = a-D glucose (C_1) → (C_4)α–D glucose

95)

NCERT, Pg. # 136

96) NCERT, Pg. # 45

97) NCERT-XI, Pg # 150

98)

NCERT-XI, Pg. # 137

99)

NCERT-XI, Pg. # 138, 139, 140, 142

100) NCERT Pg.159

101) NCERT Pg. # 160

102) NCERT Pg. # 161-164

103) NCERT-XI, Pg. # 156, 158

104) NCERT-XI, Pg. # 157

105)

NCERT-XI, Pg. # 175, 176, 177, 178

106) NCERT-XI, Pg # 248

107) NCERT-XI, Pg. # 177
Explain Question: The question asks that Ethylene initiate flowering and synchronise fruit set in which of the following.
Concept: Ethylene.
Solution: Ethylene is used to initiate flowering and for synchronising fruit set in Pineapples.
So, the correct answer is: 3.
Final answer: Option 3

108) NCERT-XI, Pg. # 245, 247, 250

109) NCERT XI Pg. # 242

110) NCERT-XI, Pg. # 26

111) NCERT-XI, Pg. # 28, Fig. 3.2(b)

112)

NCERT-XI, Pg. # 30, 32, 33, 34

113) NCERT-XI, Pg. # 27

114) NCERT-XI, Pg. No. # 30 & 32

115) NCERT-XI, Pg.# 39

116) NCERT-XI, Pg. # 11

117)

NCERT-XI, Pg. # 17, 18

118) NCERT-XI, Pg. # 20

119) NCERT-XI, Pg # 21, Fig. 2.4(a)

120)

NCERT-XI, Pg # 20, 21

121) NCERT-XI, Pg. # 15, 16

122) NCERT-XI, Pg.# 4

123) NCERT-XI, Pg. # 4, 5

124) NCERT-XI Pg # 10, Fig. 1.1

125) NCERT XI, Pg. No. # 95, 96

126) Module Pg.# 148

127)

NCERT-XI, Pg. # 94

128)

NCERT-XI, Pg. # 100

129) NCERT-XI, Pg. # 134,135

130) NCERT-XI, Page No. # 129, 133, 137, 138

131)

NCERT-XI, Pg. # 121, 122

132)

NCERT XI Pg. No. # 127

133) NCERT-XI, Pg. # 166

134)

NCERT-XI, Pg, # 123, 126

135) NCERT-XI, Pg. # 123, 124, 126,127

136) NCERT, Pg # 136

137)

NCERT, Pg. # 48

138)

NCERT, Pg. # 39

139) NCERT, Pg. # 40, 41, 45, 49

140)

NCERT, Pg. 48

141)

NCERT, Pg. # 45

142)

Module

143)

Module

144)

Module

145) NCERT-XI, Pg. # 113 (OLD)

146)

NCERT, Pg. # 83

147) NCERT Pg. # 82,83,84

148)

NCERT, Pg. # 195

149)

NCERT, Pg. # 193, 194

150)

NCERT, Pg. # 202

151) NCERT Pg. # 288

152)

NCERT, Pg. # 199, 200

153)

NCERT, Pg. # 191

154)

NCERT, Pg. # 190

155) NCERT, Pg. # 187

156) NCERT Pg # 185 (E, H)

157) NCERT Pg. # 183

158)

NCERT, Pg. # 209

159)

NCERT, Pg. # 212, 213

160) NCERT, Pg # 205-206

161) NCERT, Pg. # 291

162) NCERT, Pg. # 212

163)

NCERT, Pg. # 213

164) NCERT Pg. # 231.

165) NCERT Pg. # 231, 232

166)

NCERT Pg # 234 (18.4.1)

167) NCERT (18.3.1)

168) NCERT XI Pg. # 247

169) NCERT Pg.#246-247.

170) NCERT Pg.#242-243.

171) NCERT XI Pg. # 249, 250

172) NCERT, Pg. # 243

173)

NCERT, Pg. # 245

174) NCERT, Pg. # 243, 244

175) NCERT, Pg. # 243

176)

NCERT, Pg. # 217, 218

177) NCERT Pg. # 218

178) NCERT Pg. No. 218

179) NCERT Pg # 225

180) NCERT Pg.#225